Experiment 1: Rule Based Process

Routing custody client messages to the right desk with simple IF–THEN keyword rules.

WHY THIS EXPERIMENT

It shows how a traditional rule-based system works, so participants see exactly what fixed rules can and cannot do before moving to machine learning.

USE CASE FOR NUVAMA

Client emails and messages reach Nuvama Asset Services every day; sending each one to the right custody desk quickly cuts handling time and avoids missed deadlines.

SAMPLE DATASET: "ASSET SERVICES"

6 ready-made rules with 38 keywords, one rule per custody desk, built from the words clients actually use when they write to that desk.

The six categories (one rule per desk)

Desk	Keywords that send a message there	Why it matters to Nuvama
Corporate Actions	`dividend, bonus, rights, split, buyback, coupon, record date`	Dividends, bonus and rights entitlements must reach the client correctly and on the record date.
Settlement	`settle, pay-in, pay-out, auction, short, delivery, T+1, T+0`	A failed or short-delivered trade costs money and must be fixed the same day.
Client Reporting	`statement, report, contract note, NAV pack, ledger`	Clients' auditors and boards depend on holding statements and NAV packs on time.
Onboarding & KYC	`KYC, PAN, FATCA, signatory, registration, UBO, address`	An account cannot operate until KYC, FPI registration and signatory details are complete.
Billing	`fee, invoice, bill, charge, GST`	Fee and invoice queries affect revenue and the client relationship.
Escalations	`complaint, unacceptable, escalate, board, regulator, wrong`	An unhappy client or a regulator mention needs a senior person quickly.

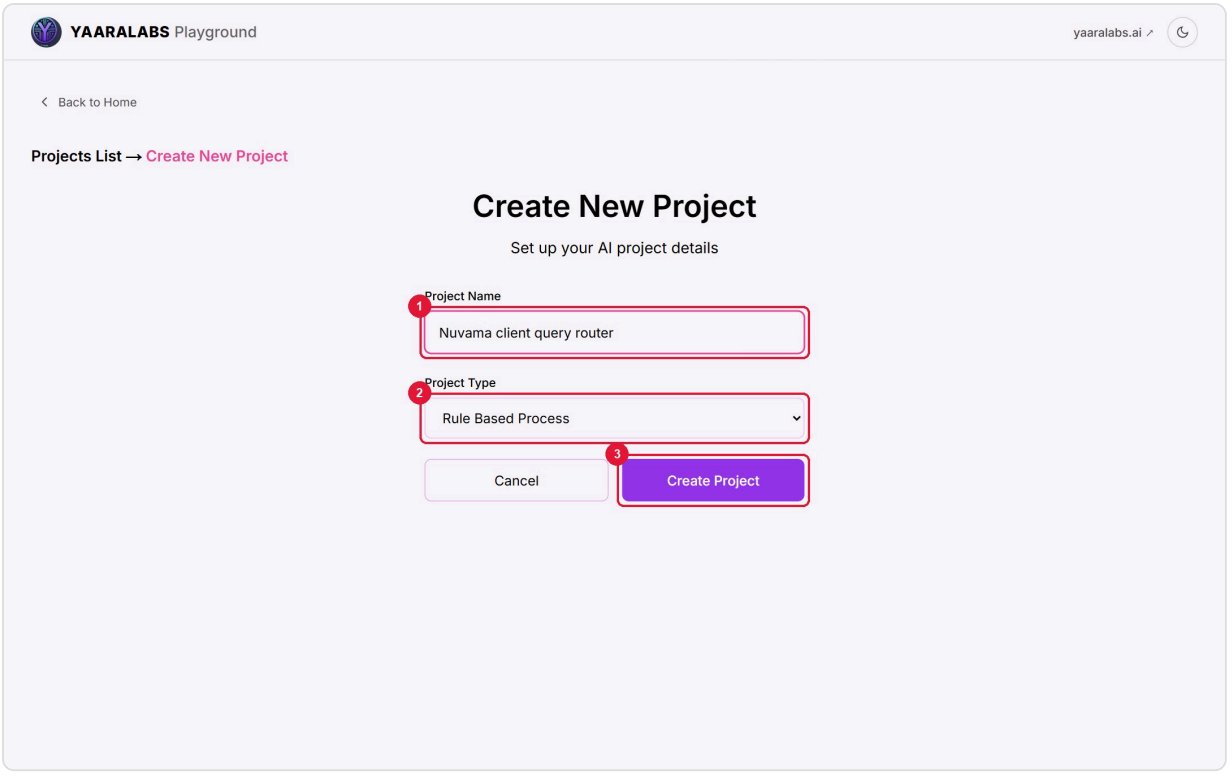
How a rule works: IF the message contains any of the rule's keywords, THEN it goes to that desk. Rules are checked top to bottom and the first match wins; if nothing matches, the bot replies "Sorry, I didn't understand. A person will reply."

Step by step

Red boxes mark what is typed or clicked; numbers show the order.

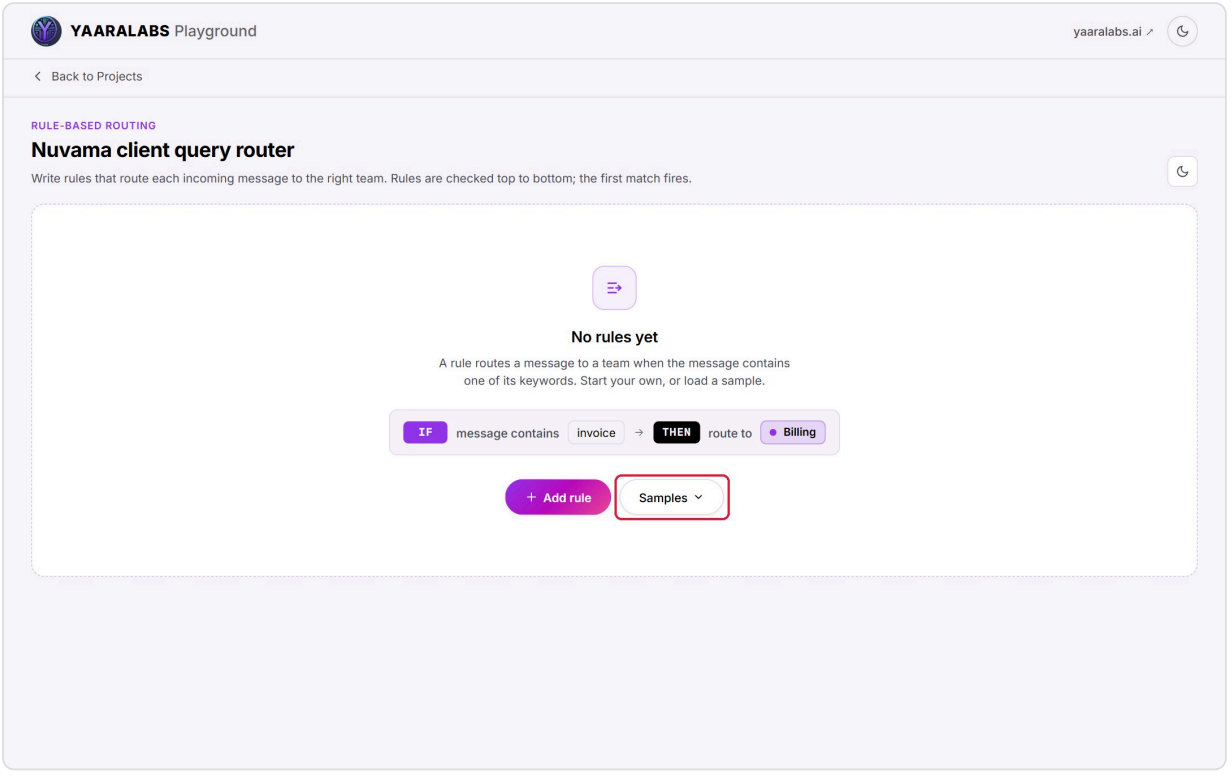
1 Create the project

Type a project name, choose “Rule Based Process” as the project type, then click Create Project.



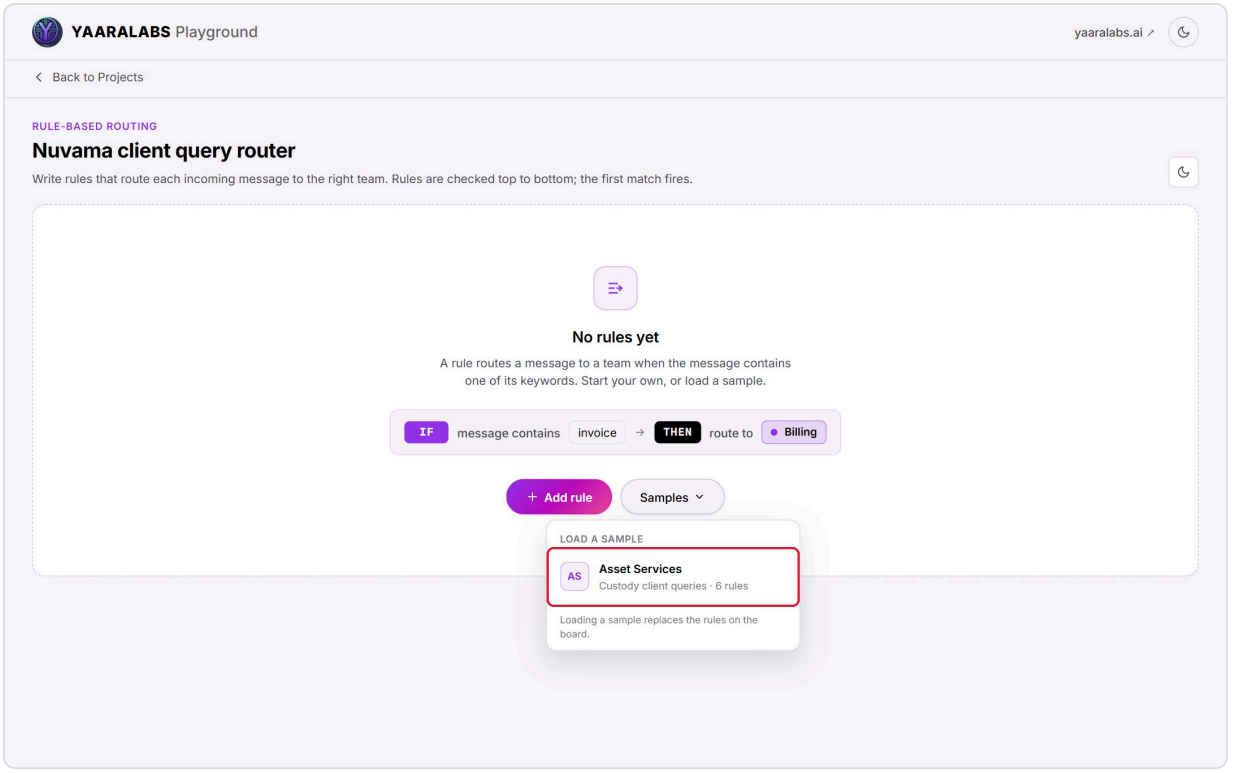
2 Open the rule board

The project opens on an empty rule board. A rule says: IF the message contains a keyword, THEN route it to a team. Open Samples.



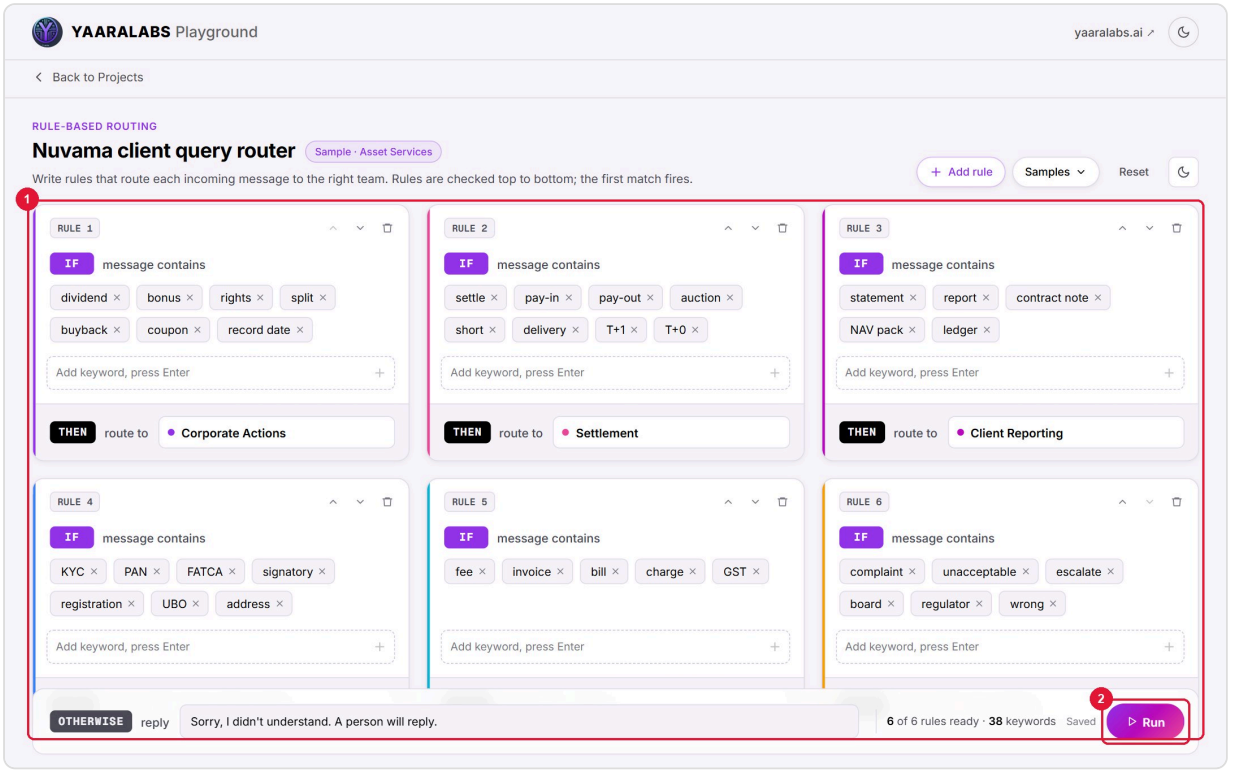
3 Load the sample rules

Choose the “Asset Services” sample: custody client queries, 6 rules.



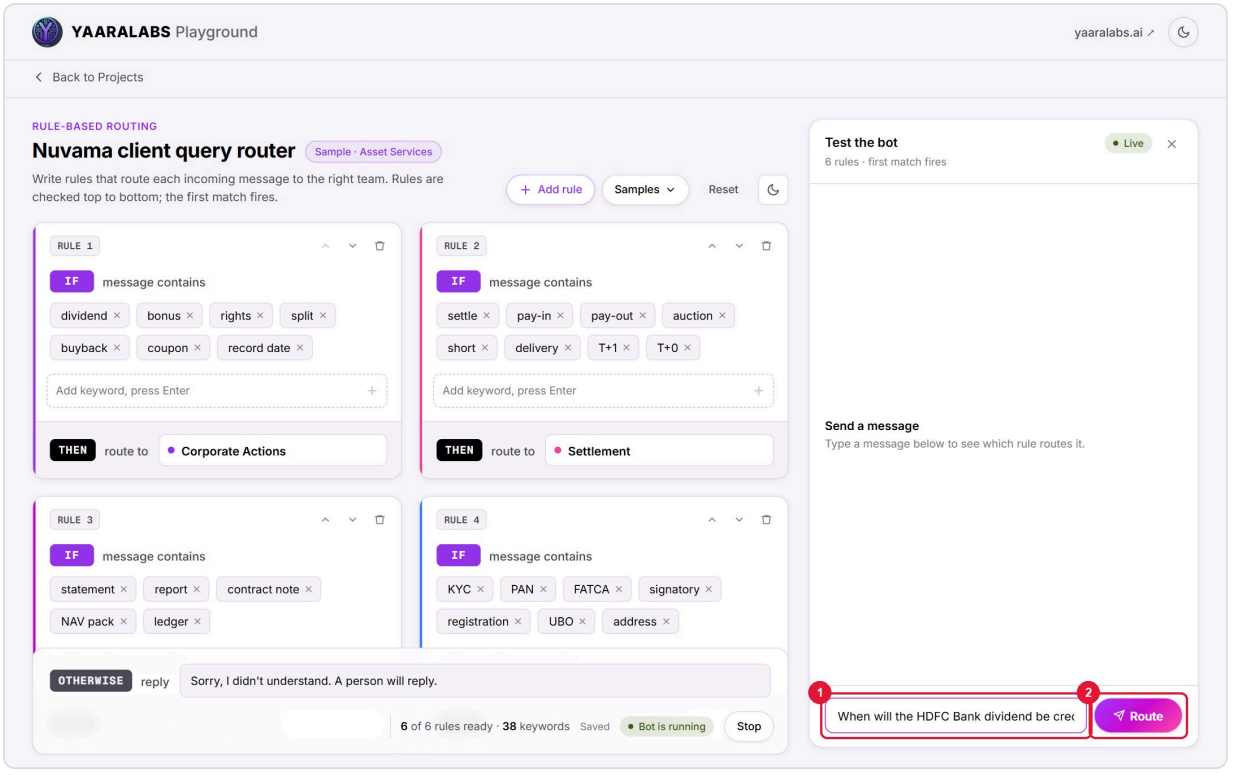
4 Review the six rules, then Run

Six rules load, one per desk, each with its keywords. Rules are checked top to bottom; the first match wins. Click Run.



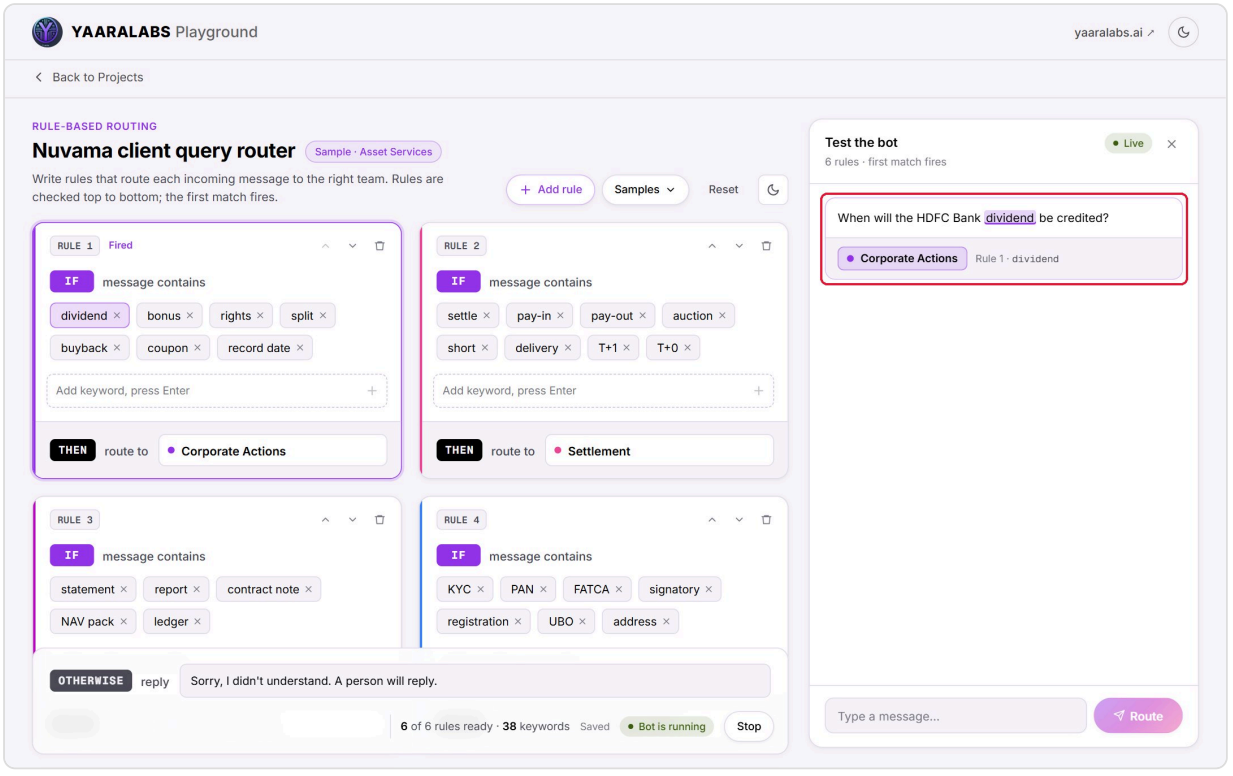
5 Test the bot with a client message

The bot is live. In the test panel on the right, type a client message and click Route.



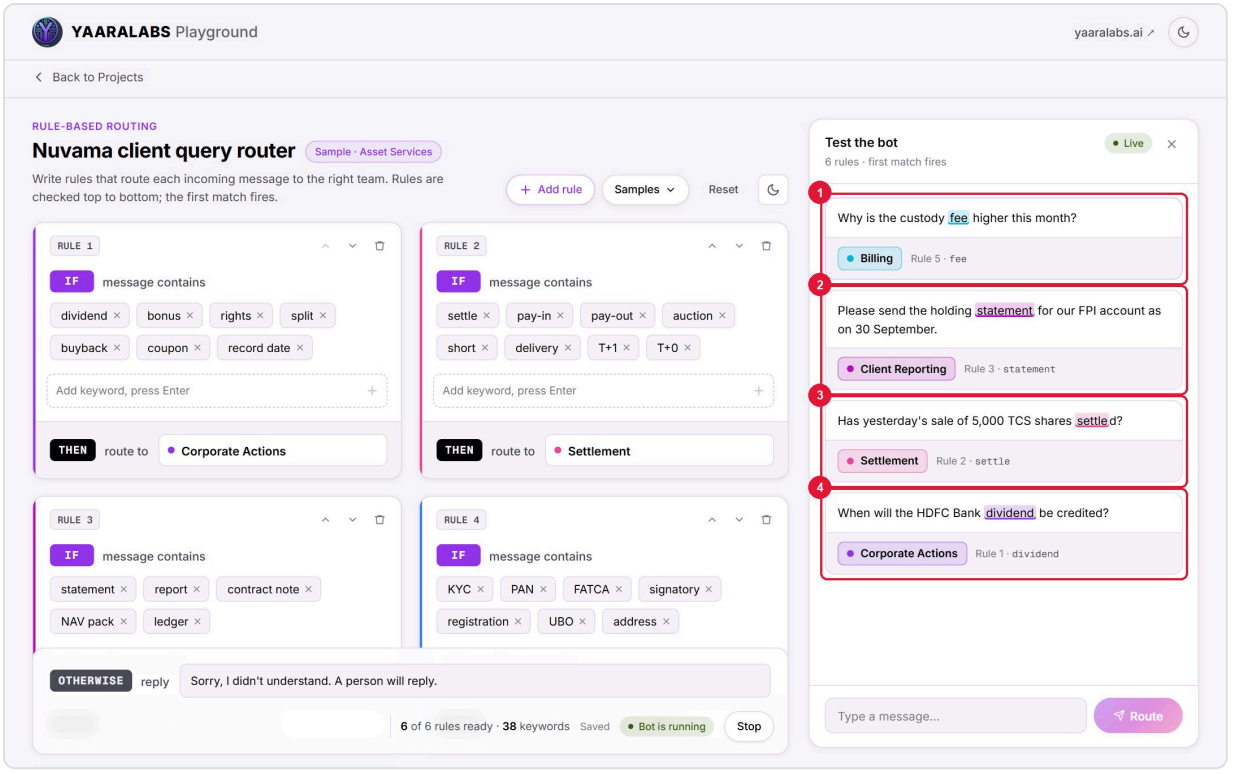
6 See which rule fired

The message goes to Corporate Actions: Rule 1 fired on the keyword “dividend”, highlighted in the message.



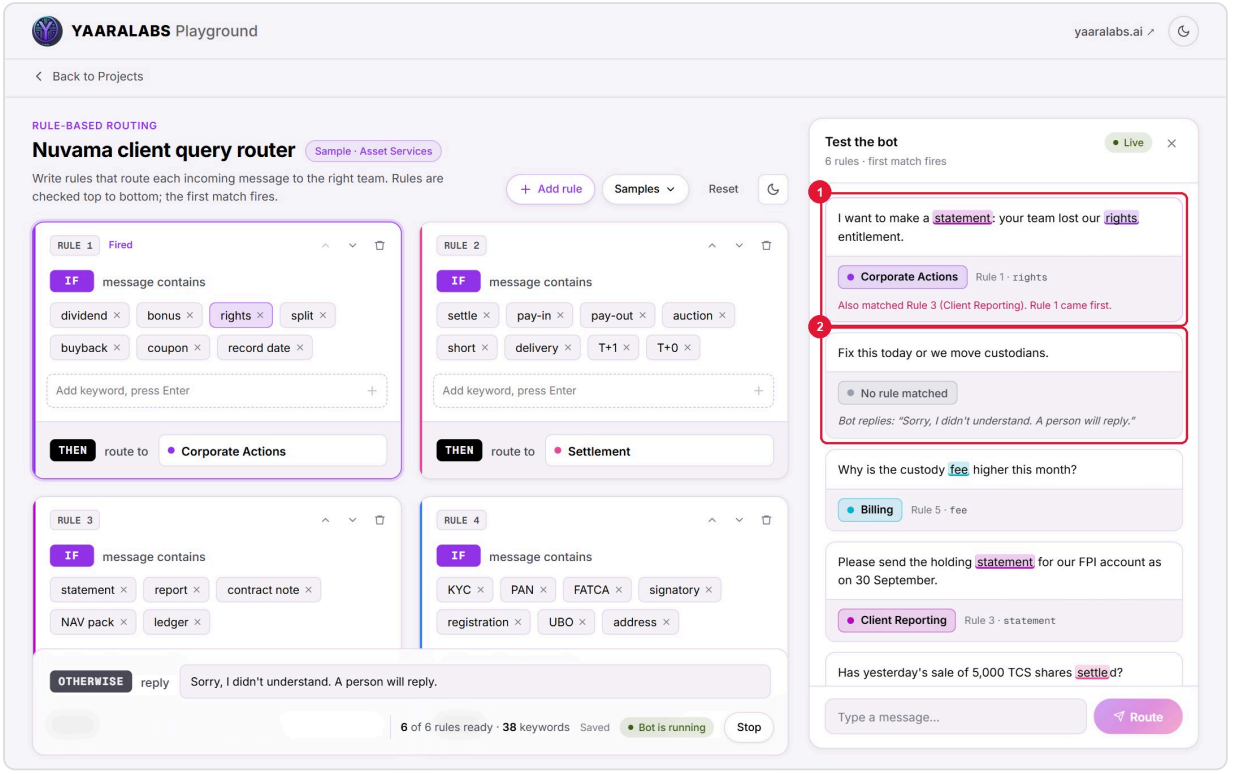
7 More messages: routed to the right desk

Four messages, four desks: Settlement (“settle”), Client Reporting (“statement”), Billing (“fee”), Corporate Actions (“dividend”).



8 Where keyword rules break

Where rules fail: a threat with no keyword gets no team, and a complaint is sent to Corporate Actions because “rights” matched first.



Test results

Client message	Bot sent it to	Keyword	Result	Comment
When will the HDFC Bank dividend be credited?	**Corporate Actions**	`dividend`	**Correct**	
Has yesterday's sale of 5,000 TCS shares settled?	**Settlement**	`settle`	**Correct**	
Please send the holding statement for our FPI account as on 30 September.	**Client Reporting**	`statement`	**Correct**	
Why is the custody fee higher this month?	**Billing**	`fee`	**Correct**	
Fix this today or we move custodians.	**No rule matched**	`none`	**Wrong**	Should be Escalations: a threat, but no keyword
I want to make a statement: your team lost our rights entitlement.	**Corporate Actions**	`rights`	**Wrong**	Should be Escalations: "rights" matched Rule 1 first

Conclusion

WHAT RULES DO WELL

- Instant and predictable: the same message always goes to the same desk.
- Explainable: the bot shows exactly which keyword and rule fired.
- No training data needed, and anyone can edit a rule.

WHERE RULES FAIL

- No keyword, no route: "Fix this today or we move custodians" reached nobody.
- Order beats meaning: a complaint went to Corporate Actions because "rights" matched first.
- Every new wording, synonym or Hinglish phrase needs another keyword by hand.

Takeaway: rules route 4 of 6 test messages correctly, but they read words, not meaning. Experiment 2 (Text Recognition) teaches a model from labelled examples so it can handle wording the rules never listed.

Experiment 2: Text Recognition

Teaching a model to sort custody client messages into six intents from labelled examples, and Logistic Regression.

WHY THIS EXPERIMENT

In Experiment 1 rules failed on messages without the right keyword; here the model learns from labelled examples, so it can handle wording no one wrote a rule for.

USE CASE FOR NUVAMA

Every client email to Nuvama Asset Services is sorted into the right intent (and desk) automatically, so it reaches the right team without someone reading and forwarding it.

SAMPLE DATASET: "NUVAMA CUSTODY CLIENT QUERIES"

36 trainer-written custody client messages, 6 for each of 6 intents (14 easy, 12 medium, 10 hard, including Hinglish and deliberate traps), plus 12 unseen test messages: 10 with a correct intent and 2 that fit none.

The six intents in the data

Label (desk)	What it covers	Example query from the data	Why it matters to Nuvama
Statement request Client Reporting desk	Asks for a statement, report, contract note, NAV pack or other record of holdings and transactions.	*"Can you resend last week's contract notes? Our auditors are asking for them."*	Auditors, boards and fund accountants need holdings and transaction records on time.
Corporate action Corporate Actions desk	Dividends, bonus, rights, splits, buybacks, mergers, record dates and entitlements, including tax on them.	*"When will the HDFC Bank dividend be credited to our account?"*	Dividends, bonus, rights and buybacks must reach the client correctly by the record date.
Settlement status Settlement desk	Trades, pay-in and pay-out, delivery, short delivery, auctions, cut-off times and settlement codes.	*"Has yesterday's sale of 5,000 TCS shares settled?"*	Unsettled or short-delivered trades cost money and need action the same day.
Fees and billing Billing desk	Custody fees, invoices, GST on invoices, charges and fee schedules.	*"We were charged twice on the September invoice."*	Wrong charges or invoices hurt revenue and trust; they go to the billing team.
Onboarding and KYC Onboarding & KYC desk	FPI registration, account opening, KYC, signatories, UBO, FATCA/CRS and address changes.	*"Any update on our FPI registration?"*	The account cannot operate until registration, KYC and signatory details are complete.
Complaint or escalation Escalations desk	The client is unhappy, threatens to move, suspects fraud or asks for a senior person. Overrides the topic.	*"This is the third failed trade this month. This is unacceptable."*	An unhappy client, a threat to leave or suspected fraud needs a senior person, whatever the topic.

How the participant categorises the data

1 · READ
Each of the 36 queries is shown unlabelled, with its full text.

2 · CHOOSE
Pick one of the 6 labels from the dropdown; the guide says what each covers.

3 · COUNT
Counters show progress; at least 5 per label are needed to train.

4 · CHECK
“Check my labels” compares with the trainer’s answers and explains each difference.

5 · GROUP
“Group it” turns each label into a box of examples: the training data.

The traps are deliberate: “Your team missed our buyback tender. We want to speak to your head of operations.” mentions a corporate action but is a complaint, because the client asks for a senior person. Labelling teaches people that the intent, not the keyword, decides the label.

The two models

LOGISTIC REGRESSION
Counts the words and word pieces in each message and learns how strongly each one points to each label. Trains in seconds, works with few examples, easy to explain.

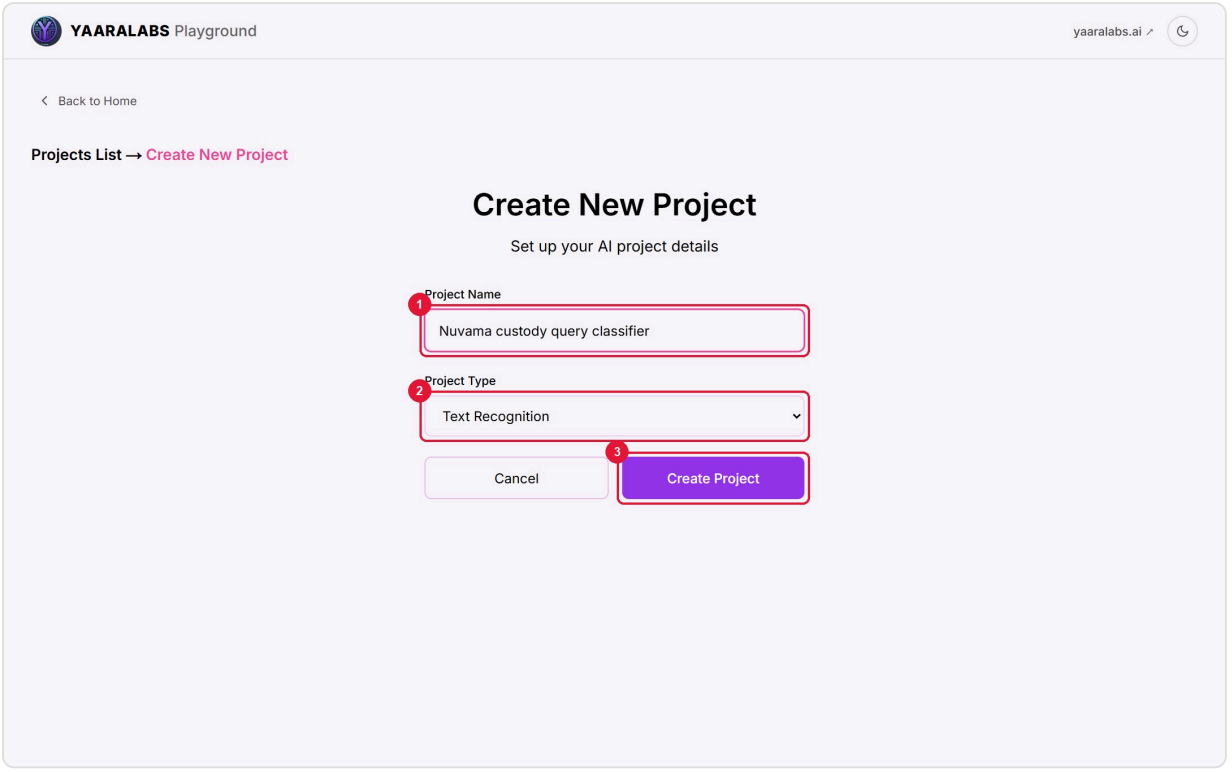
DISTILBERT (ALSO AVAILABLE)
A pretrained language model that reads the whole sentence and its context. It is the second option in the same model list; slower to train and usually needs more examples per label, so this walkthrough uses Logistic Regression.

Step by step

Red boxes mark what is typed or clicked; numbers show the order.

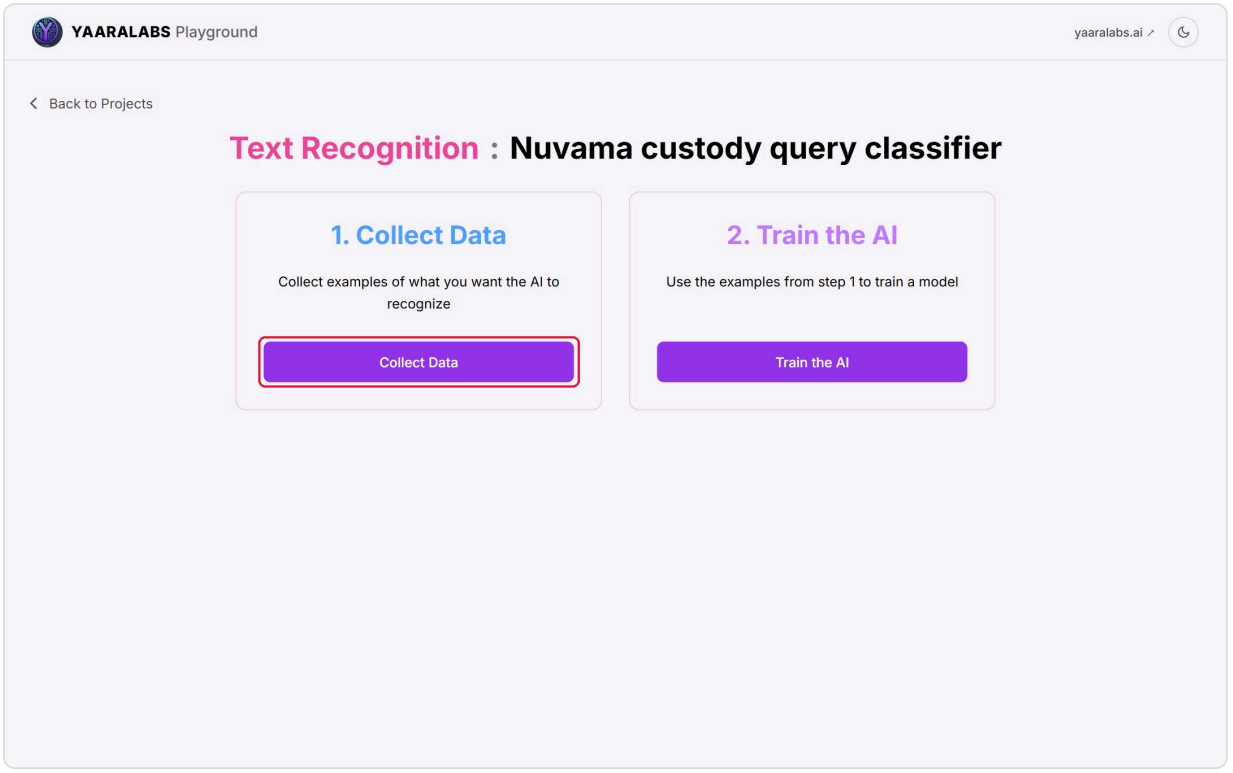
1 Create the project

Type a project name, choose “Text Recognition”, then click Create Project.



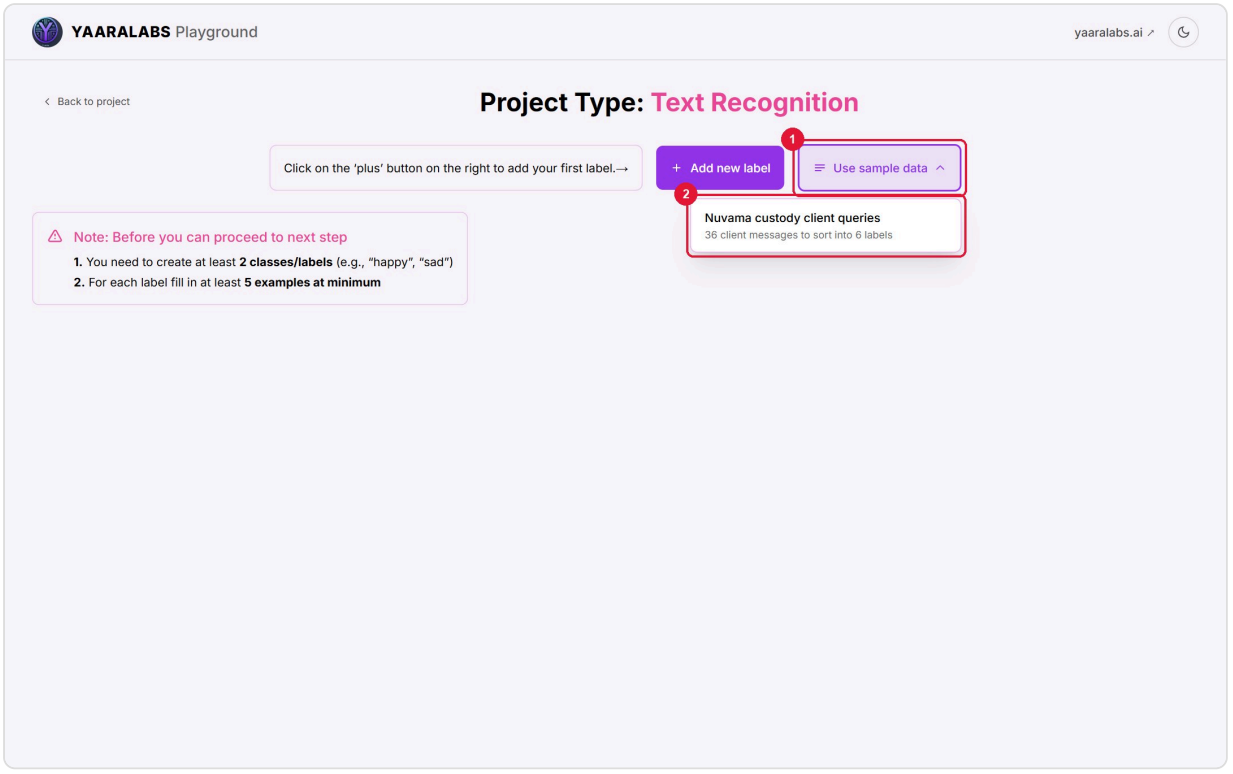
2 Open Collect Data

The project has three stages: collect data, train, use. Start with Collect Data.



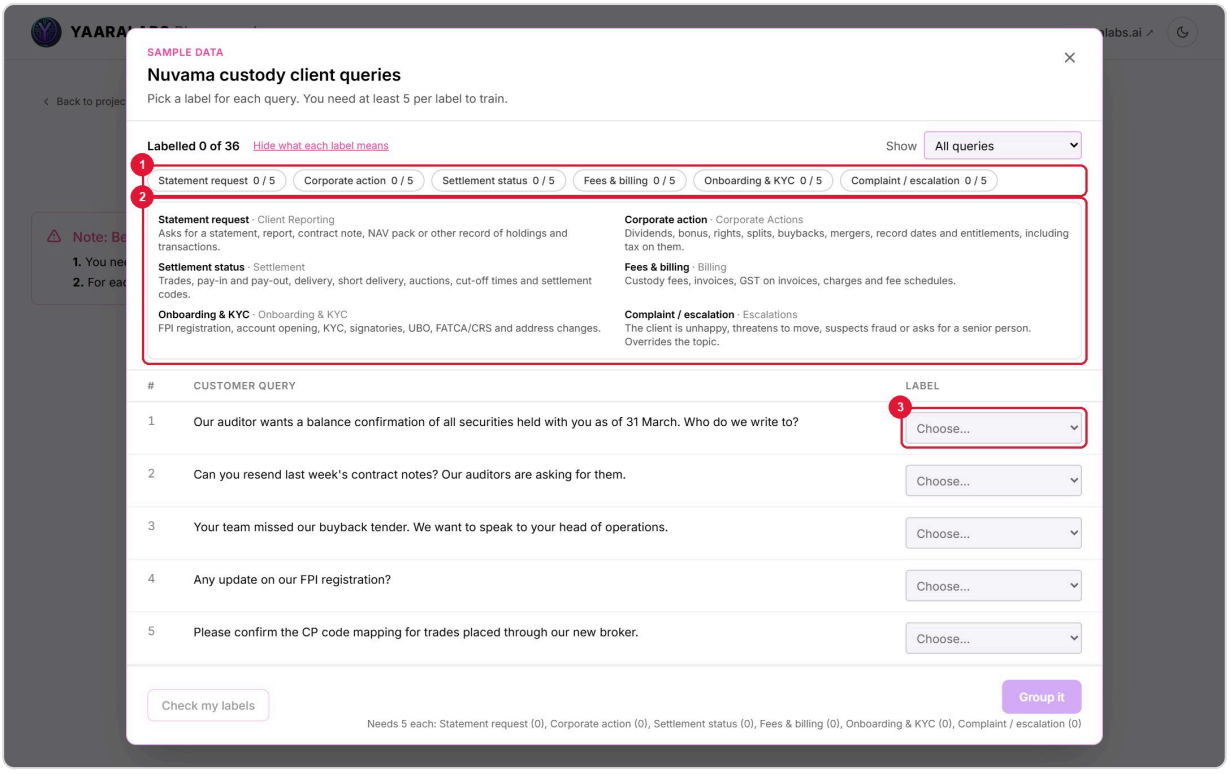
3 Load the sample data

Instead of typing labels by hand, open “Use sample data” and pick “Nuvama custody client queries”.



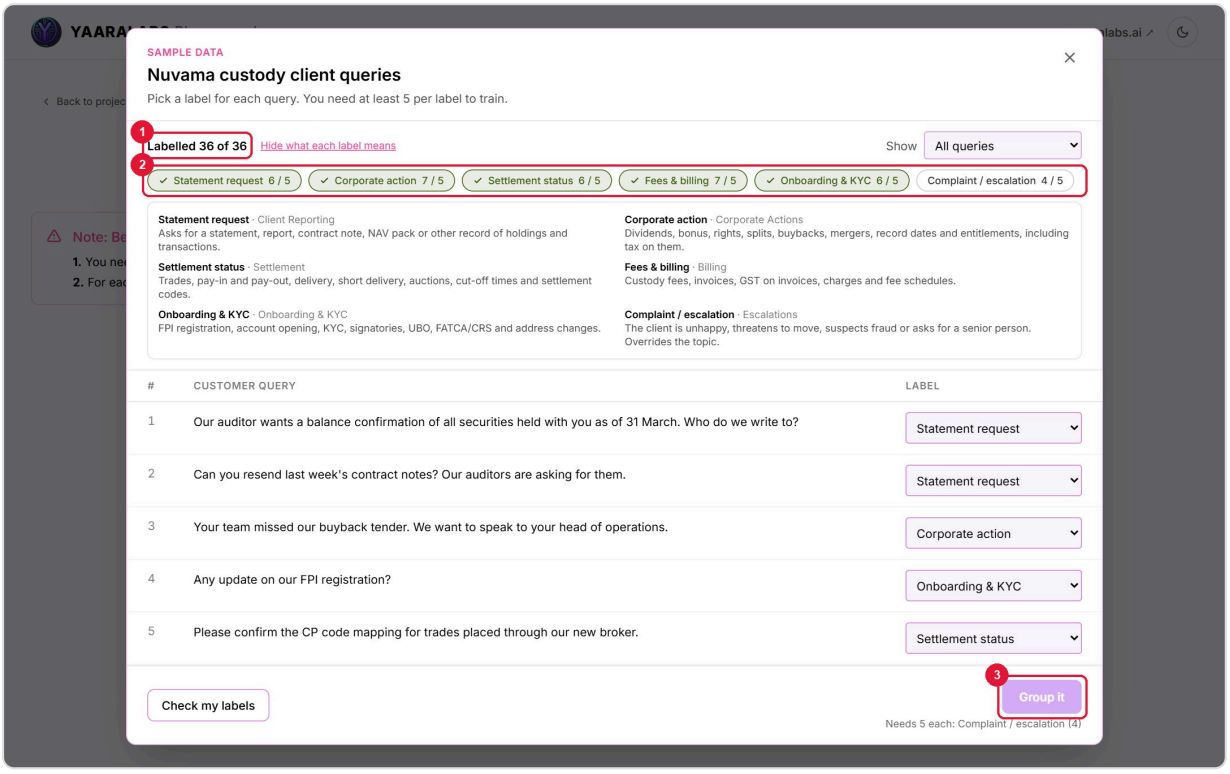
4 The labelling table

36 real-style client queries, none labelled. The six labels and what each covers are shown; the counters track 5 needed per label.



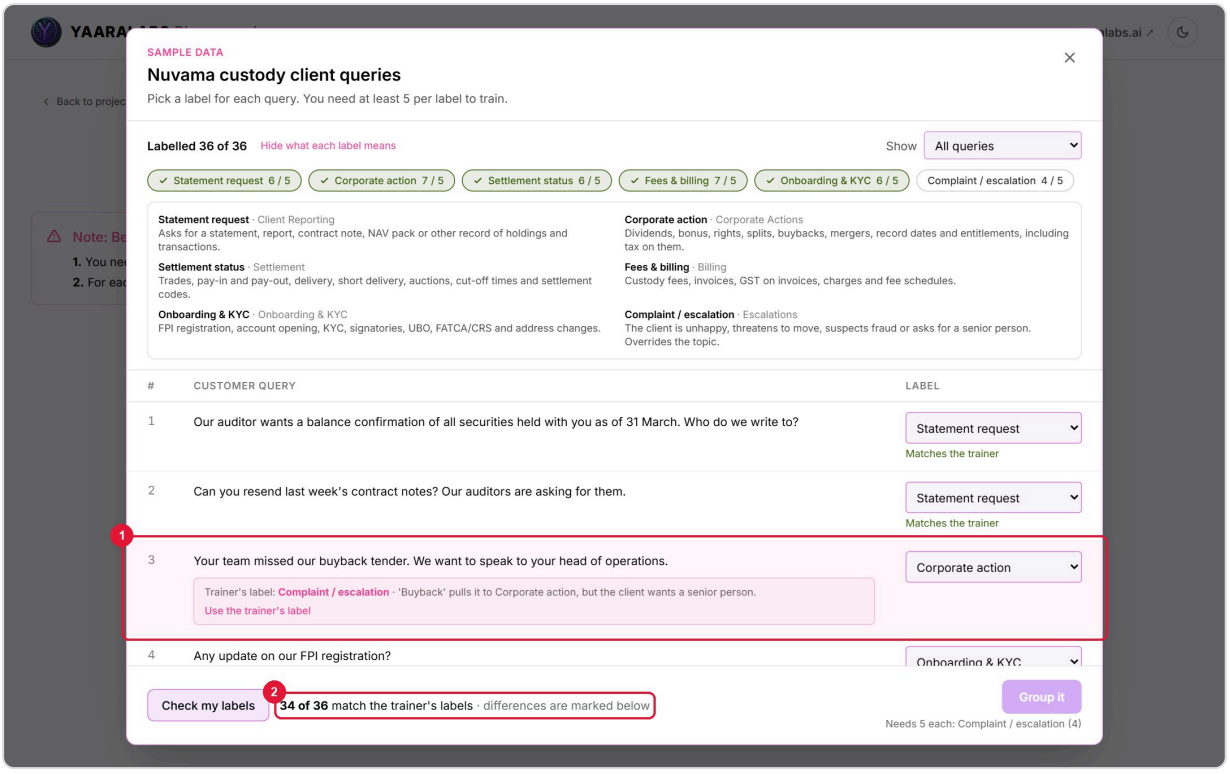
5 Label every query

Each query gets a label from its dropdown. All 36 labelled, every counter is green, so Group it is enabled.



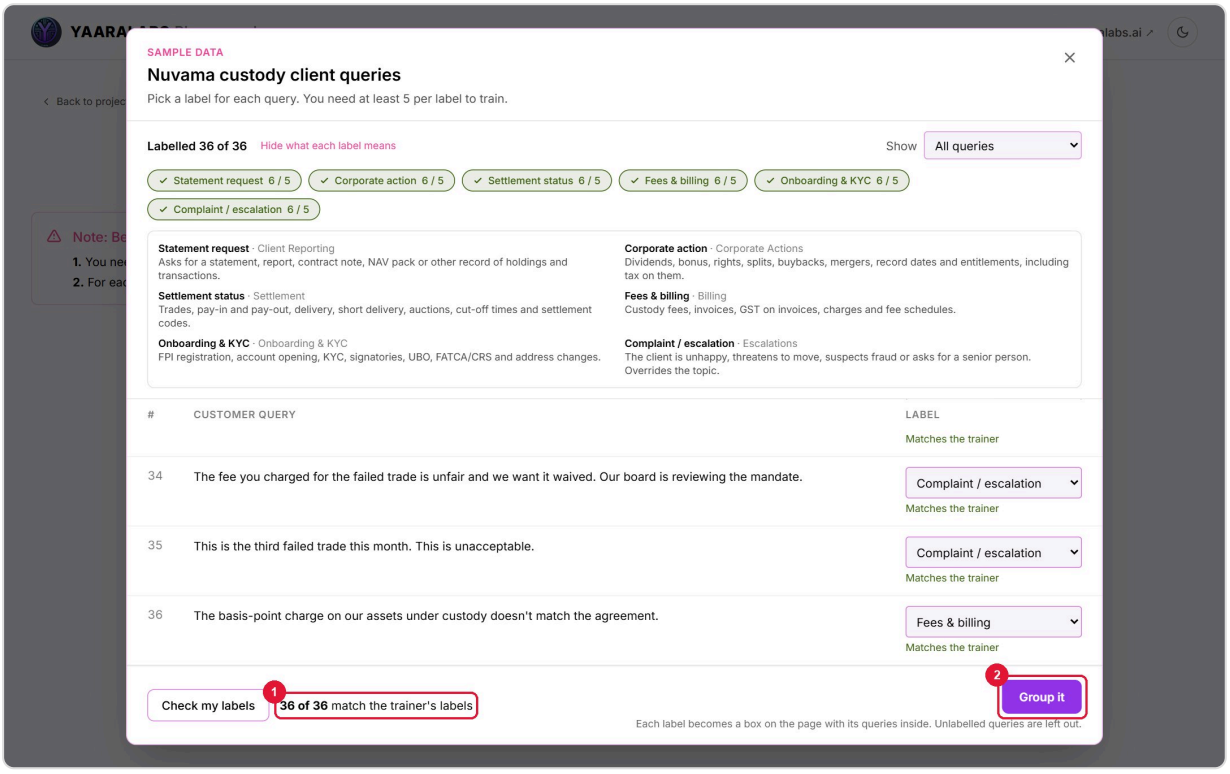
6 Check my labels

"Check my labels" compares with the trainer: 34 of 36 match. Row 3 mentions a buyback, but the client wants a senior person, so it is a complaint.



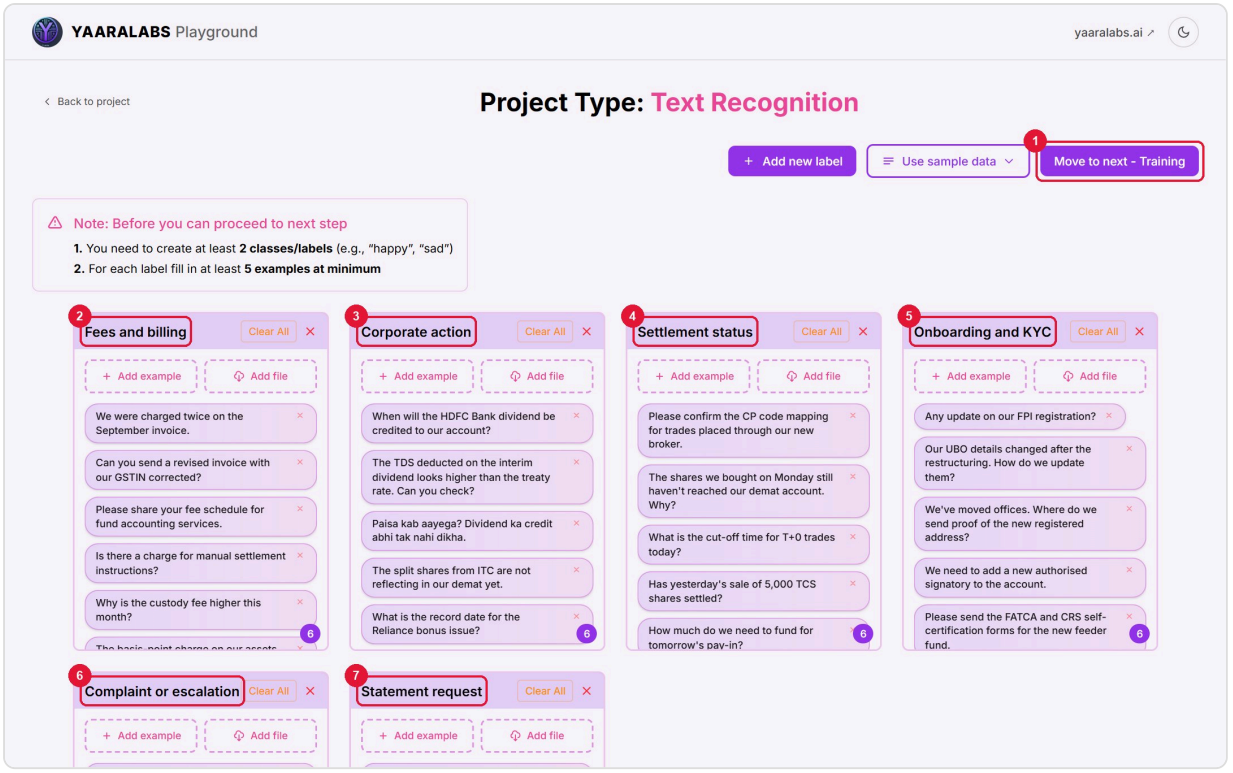
7 Fix the differences, then Group it

Accept the trainer's label on the two rows: 36 of 36 now match. Click Group it.



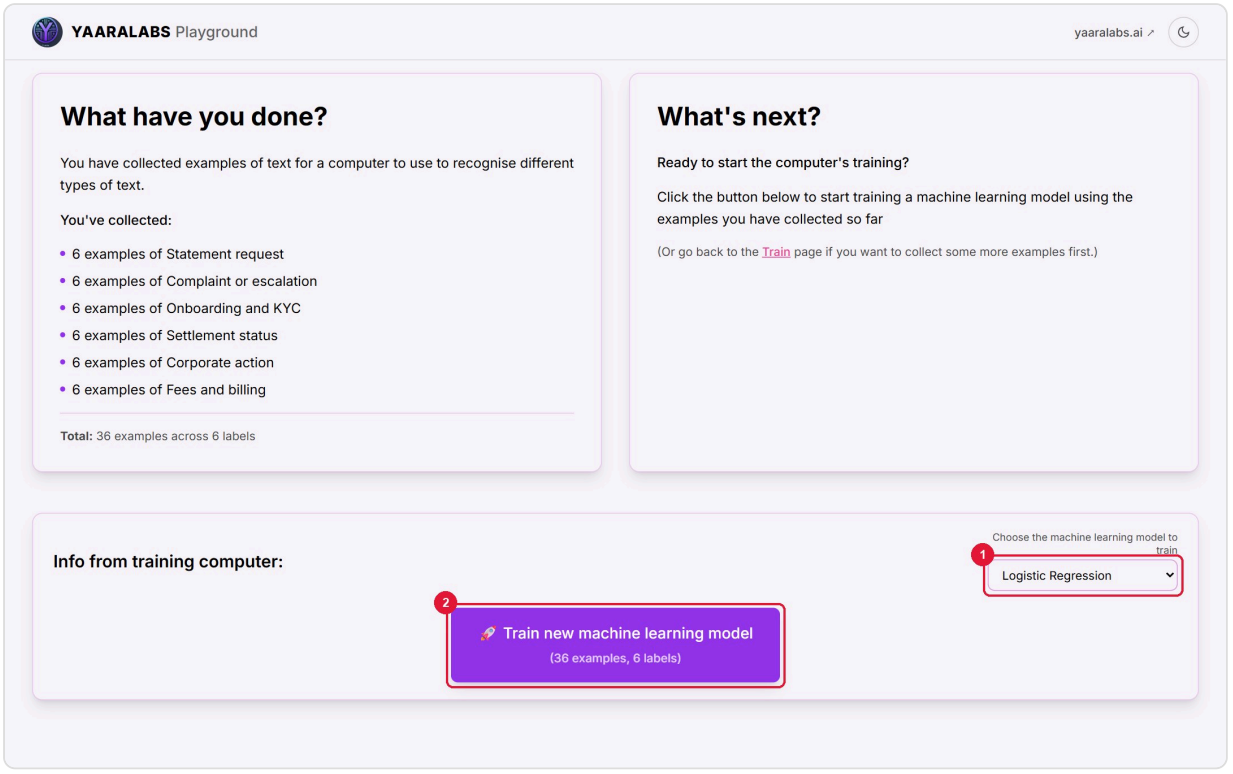
8 Each label becomes a box

Group it creates one box per label with its queries inside, just like labels added by hand. Move to Training.



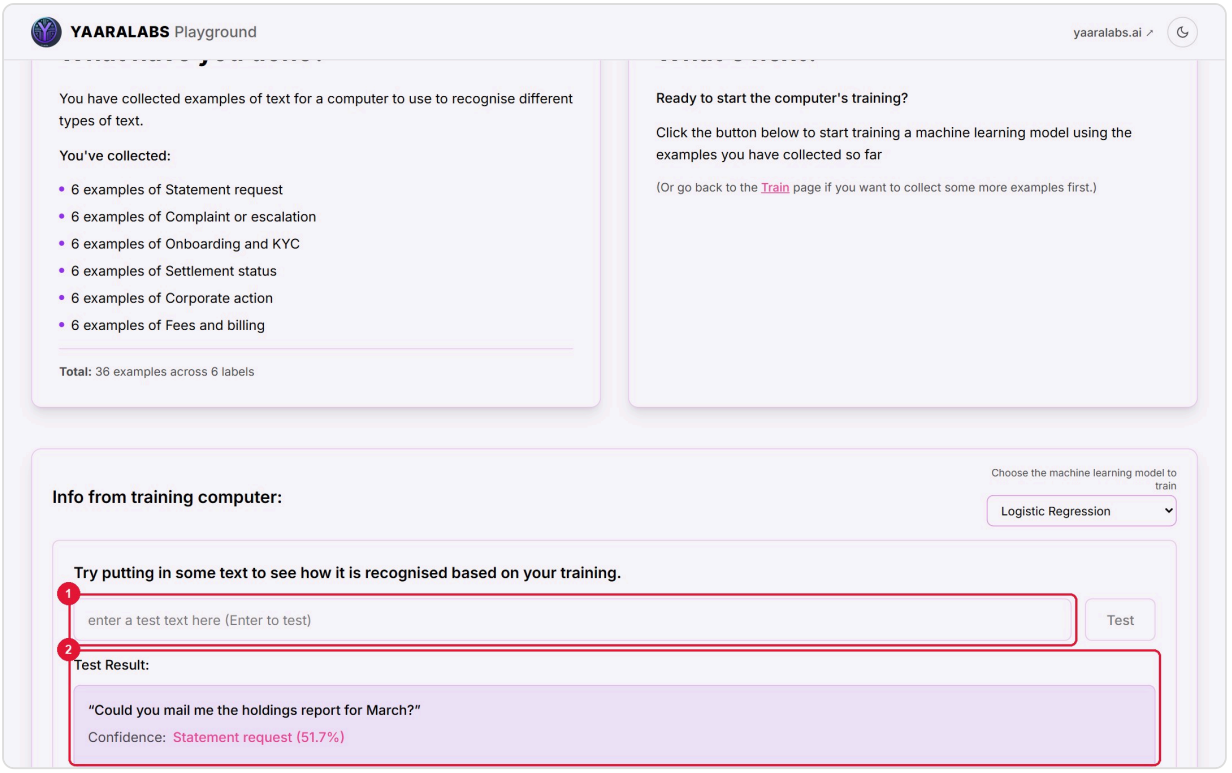
9 Choose the model and train

The model list offers Logistic Regression and DistilBERT. Choose Logistic Regression and click Train (36 examples, 6 labels).



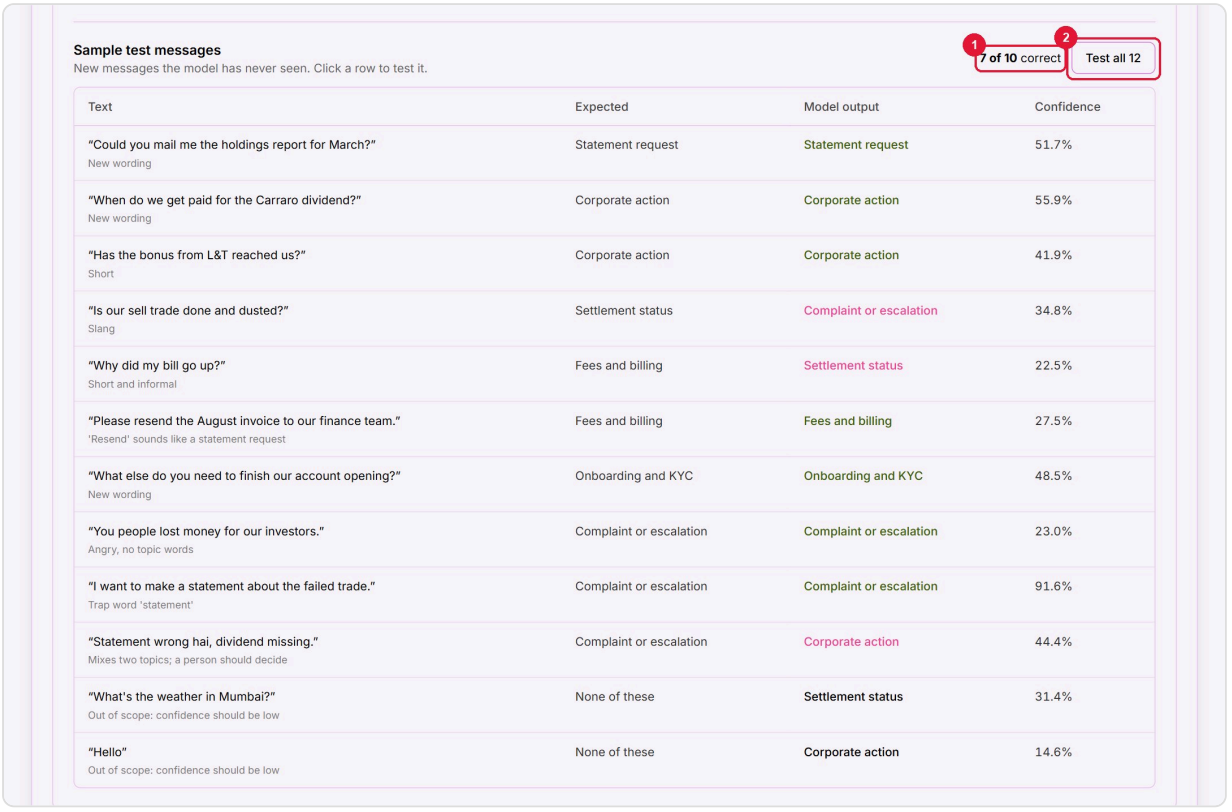
10 Try the model

Type any client message and click Test: the model returns a label and how confident it is.



11 Results on the 12 test messages

Open Test data and click Test all 12: each unseen message shows the expected label, the model's label and its confidence.



Text	Expected	Model output	Confidence
"Could you mail me the holdings report for March?" New wording	Statement request	Statement request	51.7%
"When do we get paid for the Carraro dividend?" New wording	Corporate action	Corporate action	55.9%
"Has the bonus from L&T reached us?" Short	Corporate action	Corporate action	41.9%
"Is our sell trade done and dusted?" Slang	Settlement status	Complaint or escalation	34.8%
"Why did my bill go up?" Short and informal	Fees and billing	Settlement status	22.5%
"Please resend the August invoice to our finance team." 'Resend' sounds like a statement request	Fees and billing	Fees and billing	27.5%
"What else do you need to finish our account opening?" New wording	Onboarding and KYC	Onboarding and KYC	48.5%
"You people lost money for our investors." Angry, no topic words	Complaint or escalation	Complaint or escalation	23.0%
"I want to make a statement about the failed trade." Trap word 'statement'	Complaint or escalation	Complaint or escalation	91.6%
"Statement wrong hai, dividend missing." Mixes two topics; a person should decide	Complaint or escalation	Corporate action	44.4%
"What's the weather in Mumbai?" Out of scope: confidence should be low	None of these	Settlement status	31.4%
"Hello" Out of scope: confidence should be low	None of these	Corporate action	14.6%

Results on the 12 unseen test messages (Logistic Regression)

Logistic Regression

7 of 10 in-scope messages correct

Out-of-scope messages

31%, 15% confidence (low, as it should be)

Test message	Expected	Model said	Confidence	Result
Could you mail me the holdings report for March? New wording	**Statement request**	Statement request	51.7%	Correct
When do we get paid for the Carraro dividend? New wording	**Corporate action**	Corporate action	55.9%	Correct
Has the bonus from L&T reached us? Short	**Corporate action**	Corporate action	41.9%	Correct
Is our sell trade done and dusted? Slang	**Settlement status**	Complaint or escalation	34.8%	Wrong
Why did my bill go up? Short and informal	**Fees and billing**	Settlement status	22.5%	Wrong
Please resend the August invoice to our finance team. 'Resend' sounds like a statement request	**Fees and billing**	Fees and billing	27.5%	Correct
What else do you need to finish our account opening? New wording	**Onboarding and KYC**	Onboarding and KYC	48.5%	Correct
You people lost money for our investors. Angry, no topic words	**Complaint or escalation**	Complaint or escalation	23.0%	Correct
I want to make a statement about the failed trade. Trap word 'statement'	**Complaint or escalation**	Complaint or escalation	91.6%	Correct
Statement wrong hai, dividend missing. Mixes two topics; a person should decide	**Complaint or escalation**	Corporate action	44.4%	Wrong
What's the weather in Mumbai? Out of scope: confidence should be low	**None of these**	Settlement status	31.4%	Low confidence: good
Hello Out of scope: confidence should be low	**None of these**	Corporate action	14.6%	Low confidence: good

Conclusion

WHAT THE EXPERIMENT SHOWS

- The model learned intents from 36 labelled examples, with no rules written.
- It handles new wording the training data never contained, which rules could not.
- A confidence score tells you when to let a person decide.

WHAT TO WATCH

- Six examples per intent is very little: short, slangy or mixed messages are still missed.
- Label quality matters: the “Check my labels” step catches intent-vs-keyword mistakes.
- The model always picks a label, so out-of-scope messages need a low-confidence rule.

Takeaway: from just 36 labelled messages, Logistic Regression sorted 7 of 10 unseen client messages correctly, including the trap “I want to make a statement about the failed trade” at 92% confidence, which keyword rules got wrong in Experiment 1. More labelled examples make it better; Experiment 3 shows exactly where a model goes wrong, using a confusion matrix.

Experiment 3: Confusion Matrix

Seeing exactly which way a model is wrong, and tuning it for the mistake the business cares about most.

WHY THIS EXPERIMENT

Accuracy alone hides how a model fails; the confusion matrix shows what it caught, what it missed and what it flagged by mistake.

USE CASE FOR NUVAMA

Before trusting a model to triage client messages, Nuvama must know how many urgent ones it would miss, and how many false alarms seniors would chase.

Two ways to start: why there is an "Ex 2" box

1 · MY EX 2 MODEL

- **What:** the Text Recognition model you built in Experiment 2, on your own labelled examples.
- **How:** pick the project and the "class we care about" (e.g. Complaint or escalation vs all other labels); Ex 3 retrains it on 80% of your examples and tests it on the 20% it never saw.
- **Why:** to judge your own model honestly; Ex 2's 100% was measured on data it had already seen.
- **Limit:** with ~36 examples only ~7 are tested, so scores jump in big steps.

2 · A PRE-TRAINED MODEL

- **What:** a model already trained on a 1,600-item sample: Client emails, IT service desk or Customer feedback.
- **How:** pick a dataset and it opens instantly; 320 held-back items are scored.
- **Why:** enough test data for steady scores, and every setting visibly changes the result, so it is the clearest way to learn.
- **Used below:** Client emails.

In short: the Ex 2 box tests **your own** small model; the pre-trained box teaches **how the matrix and settings work** on a large, ready dataset. Both use logistic regression and show the same matrix.

The three sample datasets and where Nuvama would use them

Dataset	What is in it	Where Nuvama uses it	The mistake that matters most
Client emails Escalate vs Routine Used in this walkthrough	Custody client emails about dividends, buybacks, settlements, statements, TDS and fees.	Client-service and custody desks: flag the emails that need a senior person now (missed corporate action, failed settlement, auditor or regulator deadline).	**Missed escalation:** an urgent client waits; a false alarm only costs a senior's time.
IT service desk Incident vs Request	Internal IT tickets on the trading terminal, order system, settlement batch, NAV engine, depository link and VPN.	IT operations: send live outages straight to the on-call team, leave access and software requests in the normal queue.	**Missed incident:** an outage near a cut-off means failed settlements or a late NAV.
Customer feedback Complaint vs Feedback	Investor messages from the app, branches and email about SIP orders, redemptions, contract notes, demat transfers and charges.	Grievance and service teams: separate complaints that must be logged and resolved from suggestions and thanks.	**Missed complaint:** SEBI resolution timelines can be breached and it may escalate to SCORES.

Each dataset has 1,600 items (about one third in the class that matters); the model learns from 1,280 and is tested on 320 it never saw. Below, Client emails is run end to end: 108 Escalate and 212 Routine emails in the test set.

How to read the matrix

Caught (TP)
Real escalations sent to Escalate: what we want.

Missed (FN)
Real escalations left as Routine: the costly mistake, an urgent client waits.

False alarm (FP)
Routine emails sent to Escalate: wastes a senior person's time.

Correct (TN)
Routine emails left as Routine.

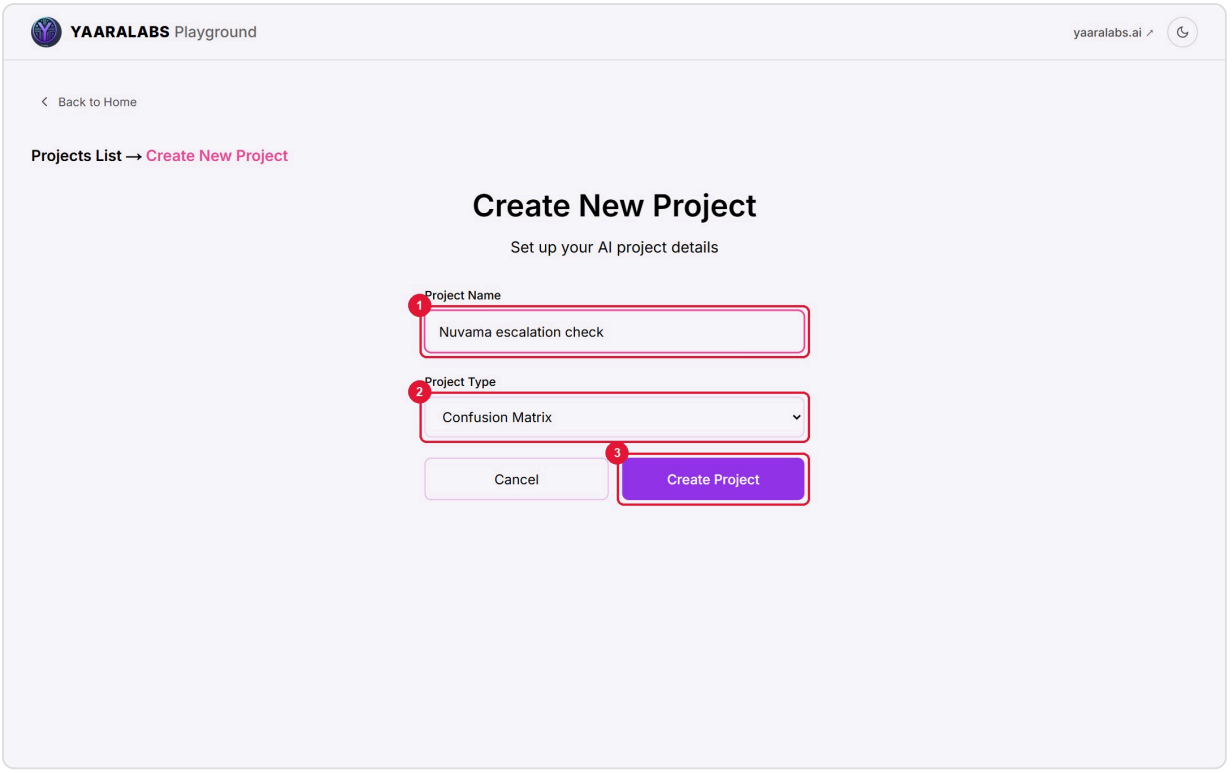
Accuracy = how often it is right overall · **Precision** = when it says Escalate, how often it is right · **Recall** = of all real escalations, how many it caught · **F1** = one score balancing precision and recall.

Step by step

Red boxes mark what is clicked or what to read; numbers show the order.

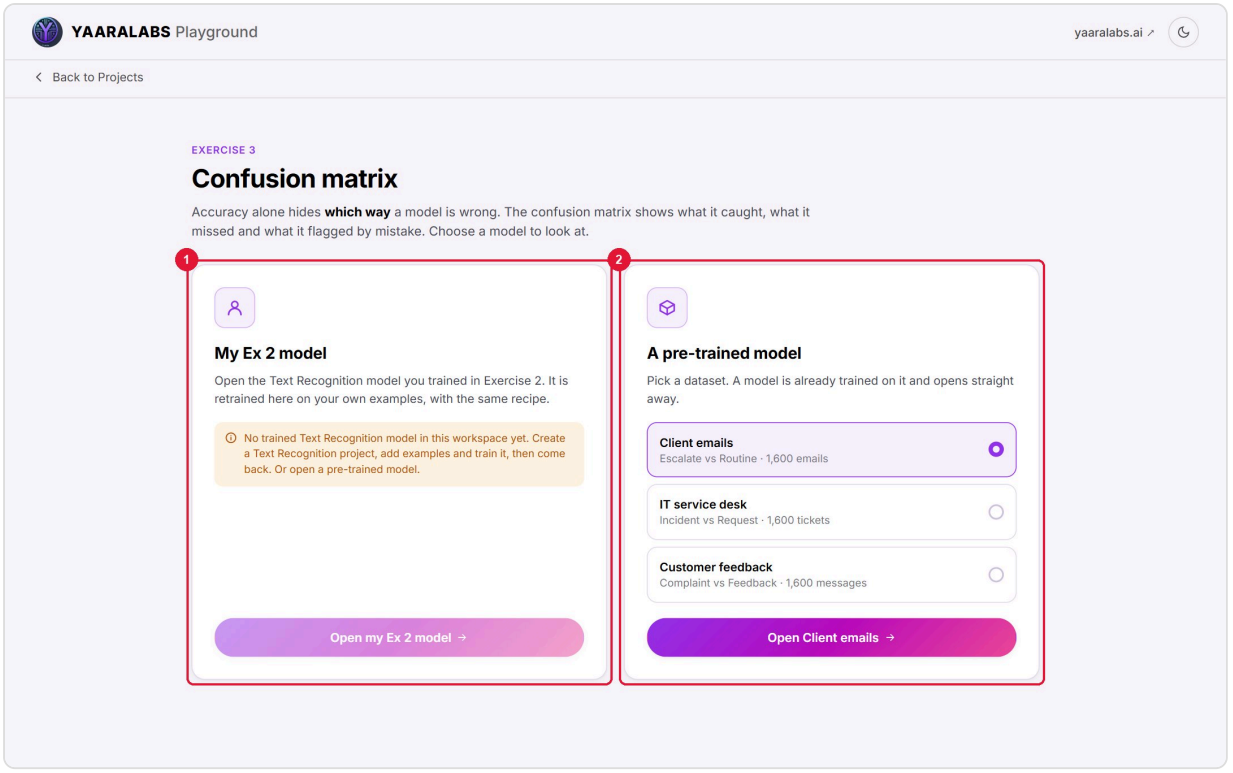
1 Create the project

Type a project name, choose "Confusion Matrix", then click Create Project.



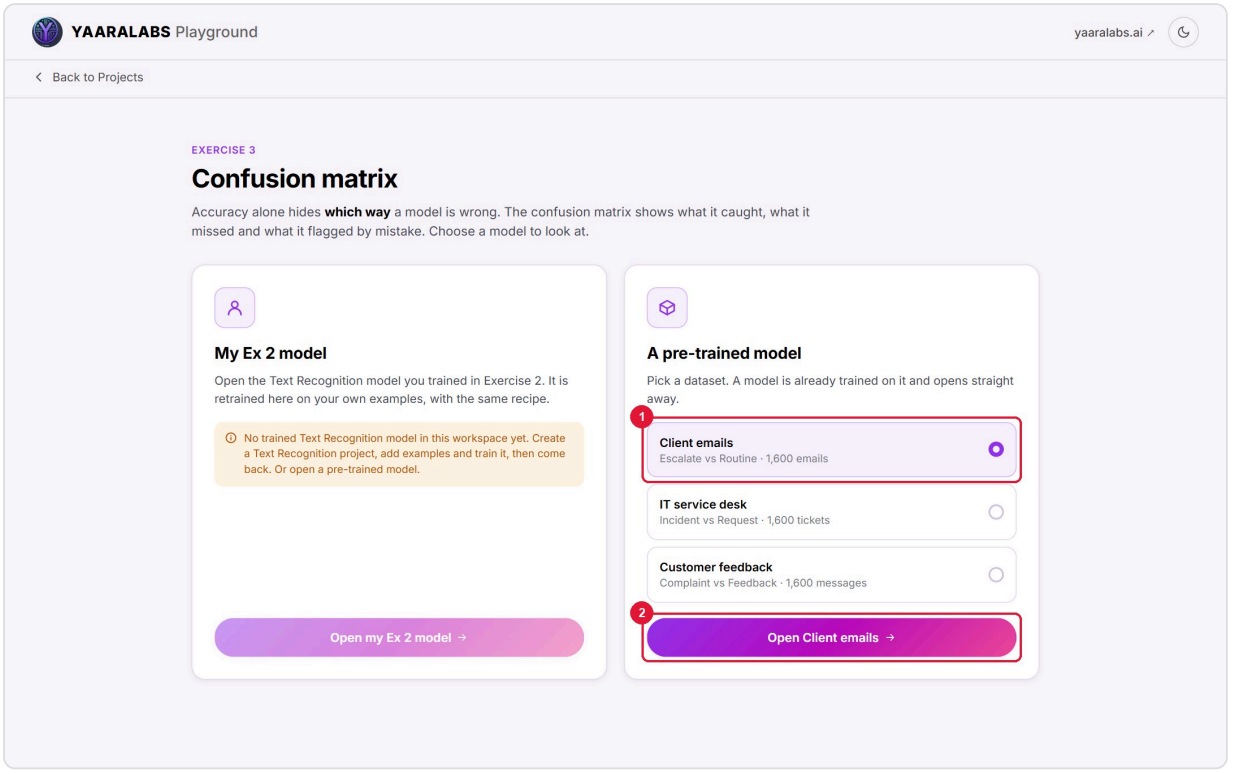
2 Choose where the model comes from

Two ways in: (1) your own model from Experiment 2, or (2) a model already trained on a sample dataset.



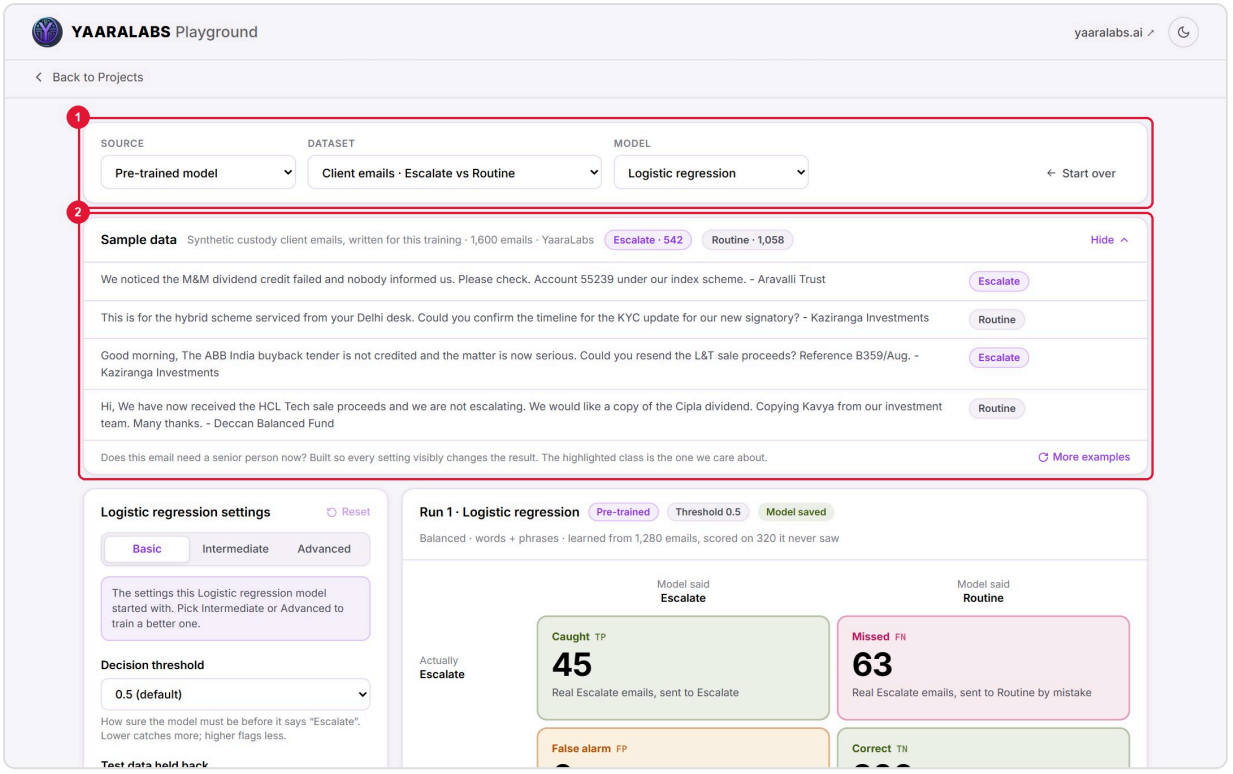
3 Open the Client emails model

Pick the “Client emails” sample (1,600 emails: Escalate vs Routine) and open it.



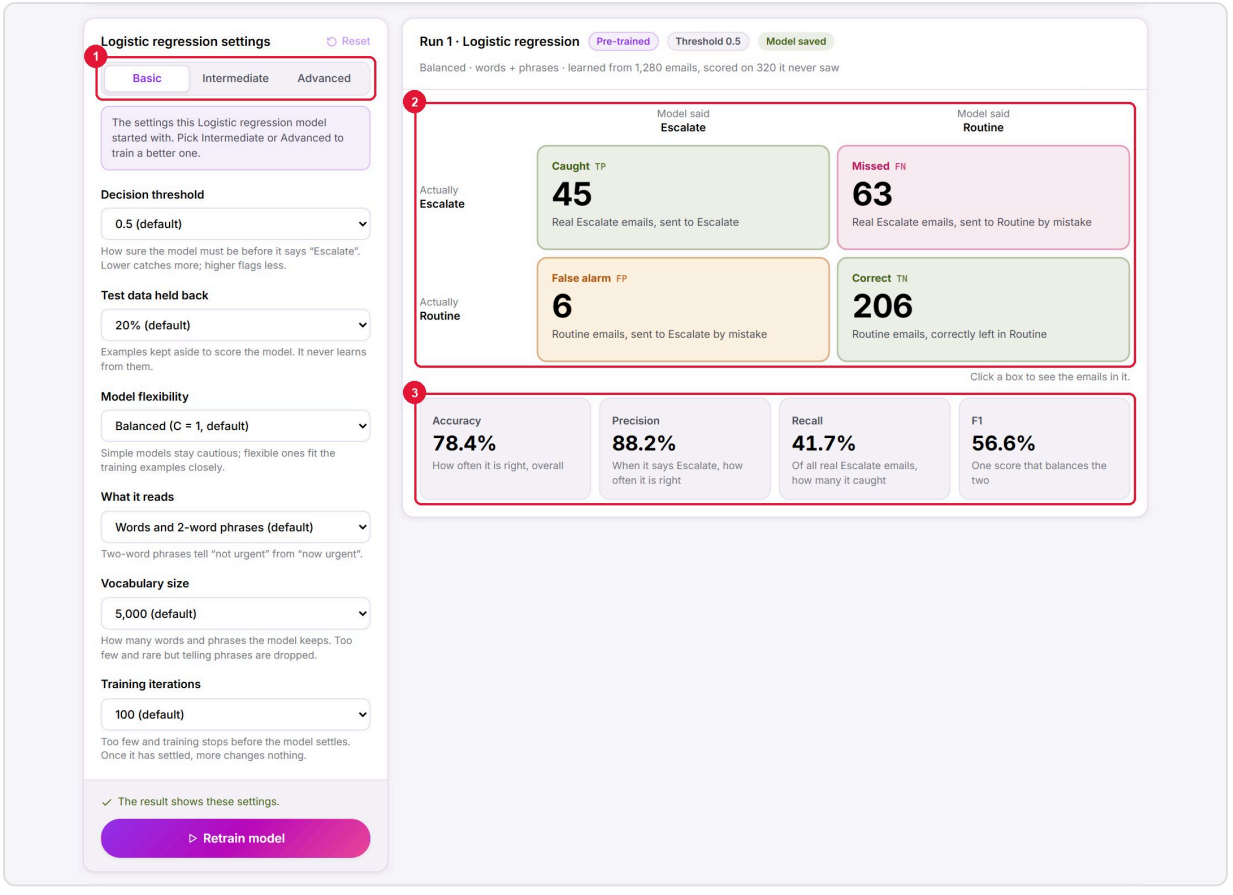
4 The workspace

The workspace shows the source, dataset and model (1), and a few real rows of the data with their true class (2).



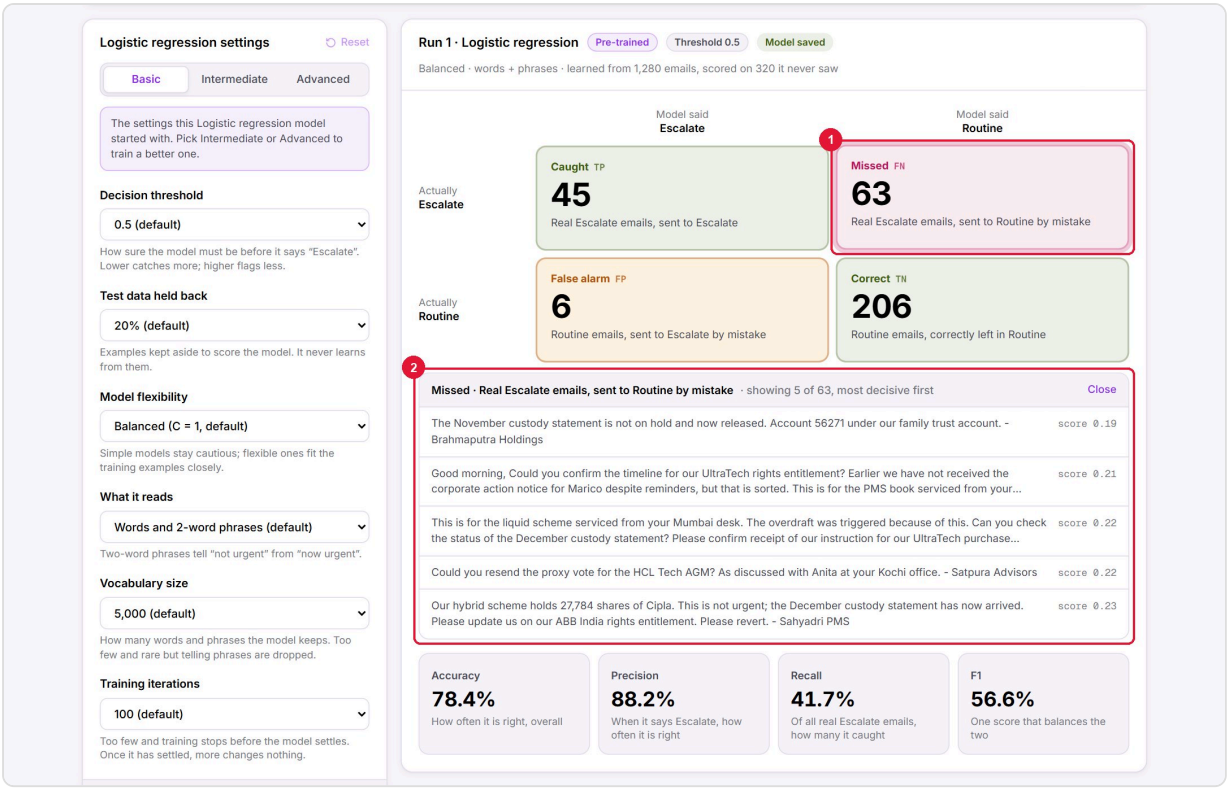
5 Run 1: the confusion matrix

Run 1 is the starting model (Basic). The matrix splits 320 test emails it never saw into four boxes; the four scores sit below.



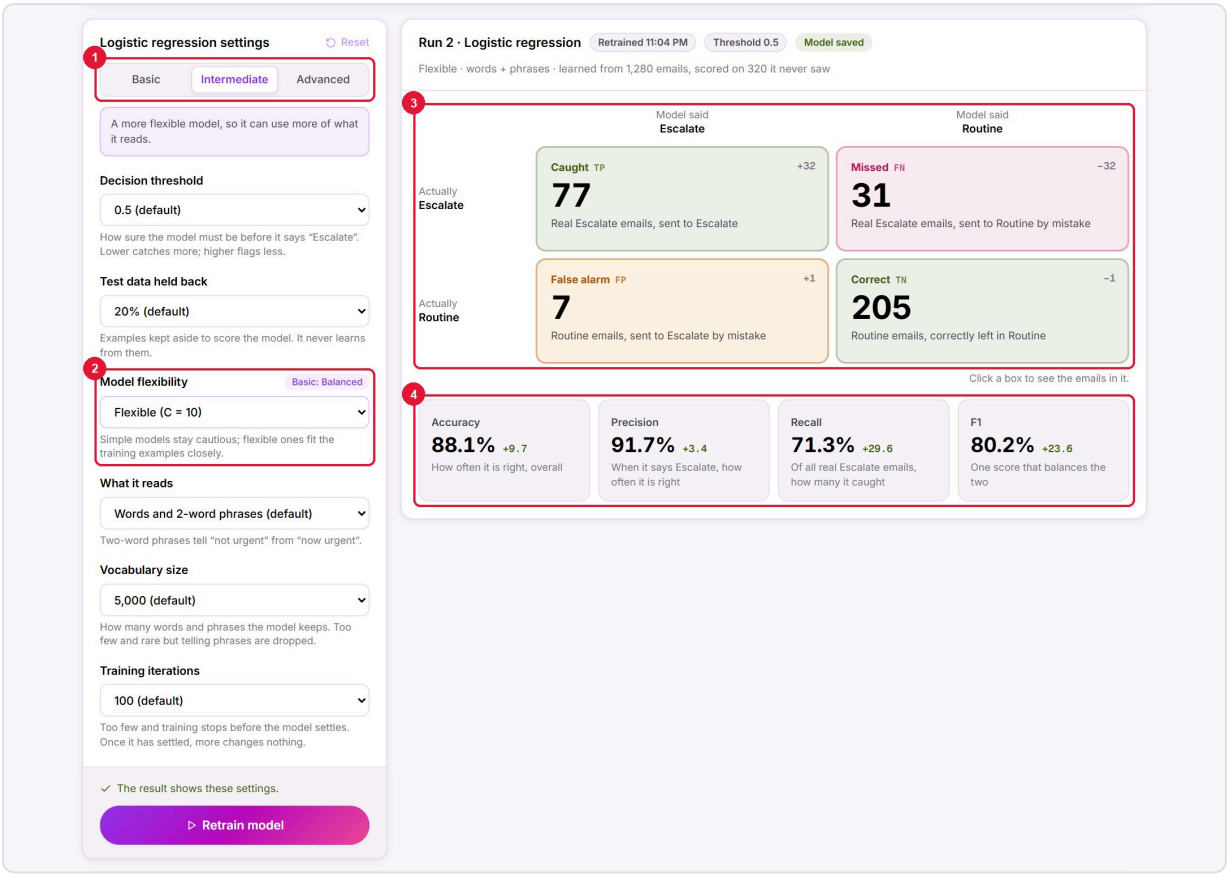
6 Look inside a box

Click a box to read the emails in it. “Missed” lists real escalations the model let through as routine.



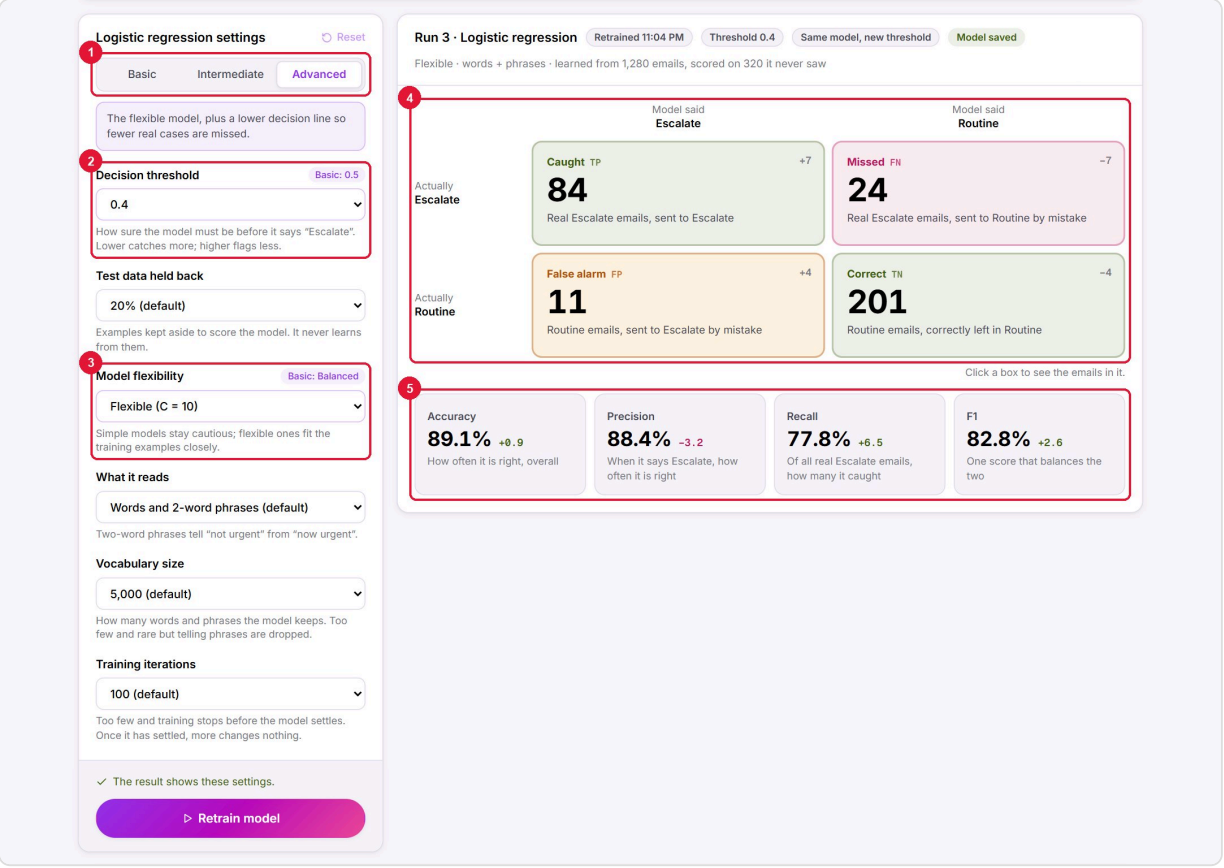
7 Intermediate: a better model in one click

Click Intermediate: the settings change (marked “Basic: …”) and the model retrains at once. Fewer misses, higher scores.



8 Advanced: tune the decision line

Click Advanced: the decision line is also lowered, so even more escalations are caught, at the cost of a few more false alarms.



9 Compare the runs

Every run is kept. Compare runs shows how each change moved Caught, Missed, False alarms and the scores.



Results: three runs on the same 320 test emails

Run	What changed	Caught	Missed	False alarms	Accuracy	Precision	Recall	F1
1	Starting model (Basic): balanced flexibility, threshold 0.5	45	**63**	6	78.4%	88.2%	**41.7%**	56.6%
2	Intermediate: model flexibility raised (C 1 → 10)	77	**31**	7	88.1%	91.7%	**71.3%**	80.2%
3	Advanced: same model, decision threshold lowered 0.5 → 0.4	84	**24**	11	89.1%	88.4%	**77.8%**	82.8%

Conclusion

WHAT THE MATRIX REVEALED

- Run 1 looked fine at 78.4% accuracy, yet missed 63 of 108 real escalations: accuracy hid it.
- A more flexible model (Intermediate) halved the misses.
- Lowering the decision line (Advanced) caught more escalations for a few extra false alarms.

THE BUSINESS DECISION

- For escalations, a miss costs far more than a false alarm, so **recall** matters most.
- The threshold is a business choice: how many false alarms seniors will accept to miss fewer urgent clients.
- Every run is saved, so the choice can be explained and reviewed.

Takeaway: from Run 1 to Run 3, missed escalations fell from 63 to 24 and recall rose from 41.7% to 77.8%, while false alarms grew only from 6 to 11. The confusion matrix turns “is the model good?” into “which mistakes can we live with?”.

Experiment 4: Image Recognition

Teaching a model to recognise what kind of document a client has sent, from just five example images per type.

WHY THIS EXPERIMENT

Experiments 1–3 worked with text; here the model learns from pictures, showing that the same "learn from labelled examples" idea works for scanned documents too.

USE CASE FOR NUVAMA

Clients send scans and phone photos of KYC forms, exchange notices, invoices and cheques; recognising the type automatically sends each to the right team and flags unreadable scans.

DATASET: "NUVAMA DOCUMENT IMAGES"

4 document types, 5 training images each (20 in total), plus 12 test images from documents never used in training: per type, 1 clean and 2 deliberately hard (blur, low light, phone-photo angle, cropped, heavy compression). Blank forms and public filings only; no personal data.

The four document types

Example	Type	Where the images came from	Who handles it at Nuvama
	**KYC** Blank KYC / demat account-opening forms (individual and non-individual)	Public blank forms from 5 different brokers and banks (Sharekhan, Motilal Oswal, JKB Financial Services, Union Bank, HSBC MF)	Onboarding & KYC team: new-account and KYC-update packs
	**Corporate action** Stock-exchange notices: dividend, bonus, buyback, split record dates	Public NSE/company filings from 5 companies (Repco Home, Ashok Leyland, eClerx, Info Edge, Remus), cropped above the signature	Corporate Actions desk: entitlements to process before the record date
	**Invoice** Business invoices, each with a different layout	FATURA dataset (CC BY 4.0), 5 different templates	Billing / accounts payable: invoices to check and pay
	**Cheque** Indian-style specimen cheques (CTS-2010 layout)	Synthetic: fictional banks, payees and numbers, marked SPECIMEN	Payments operations: cheques to verify and clear

The model: MobileNetV2 with transfer learning

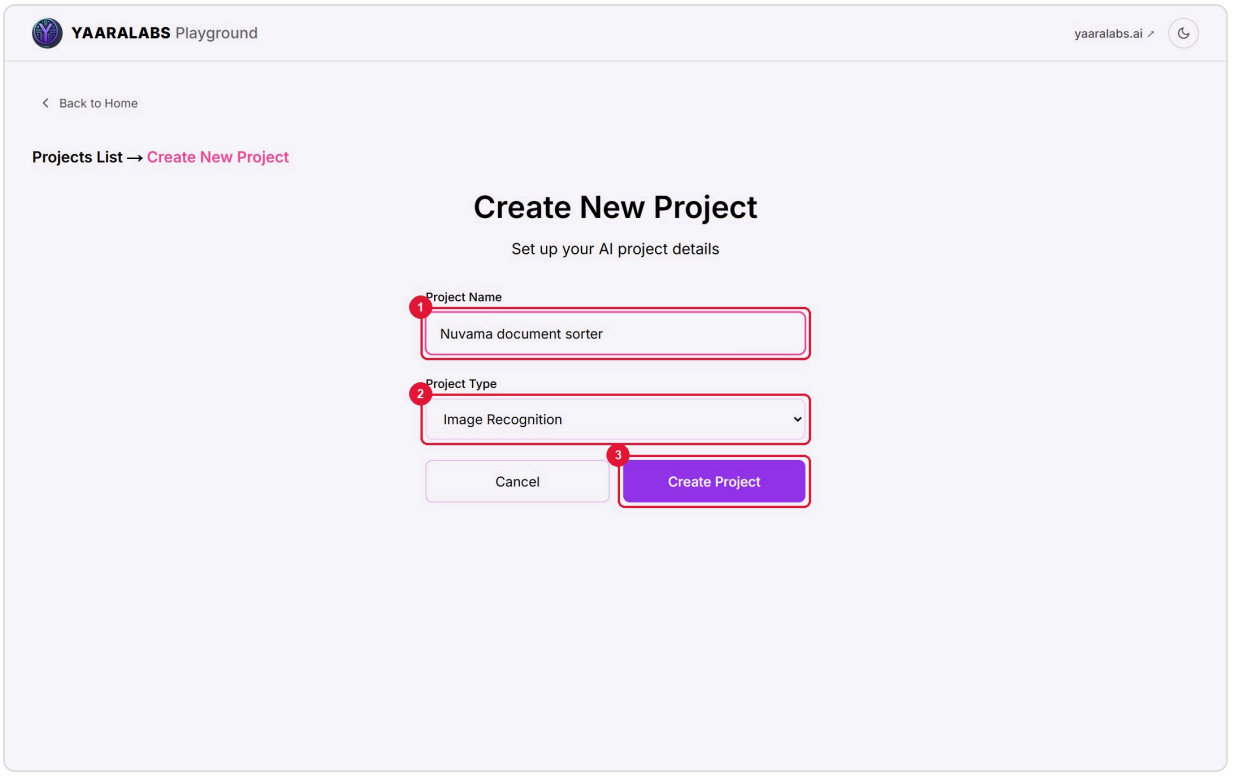
MobileNetV2 has already learned to see shapes, lines, text blocks and layouts from over a million everyday photos. We keep that knowledge and train only its last step on our 20 document images. That is why 5 examples per type are enough to learn the difference between a form, a notice, an invoice and a cheque.

Step by step

Red boxes mark what is typed or clicked; numbers show the order.

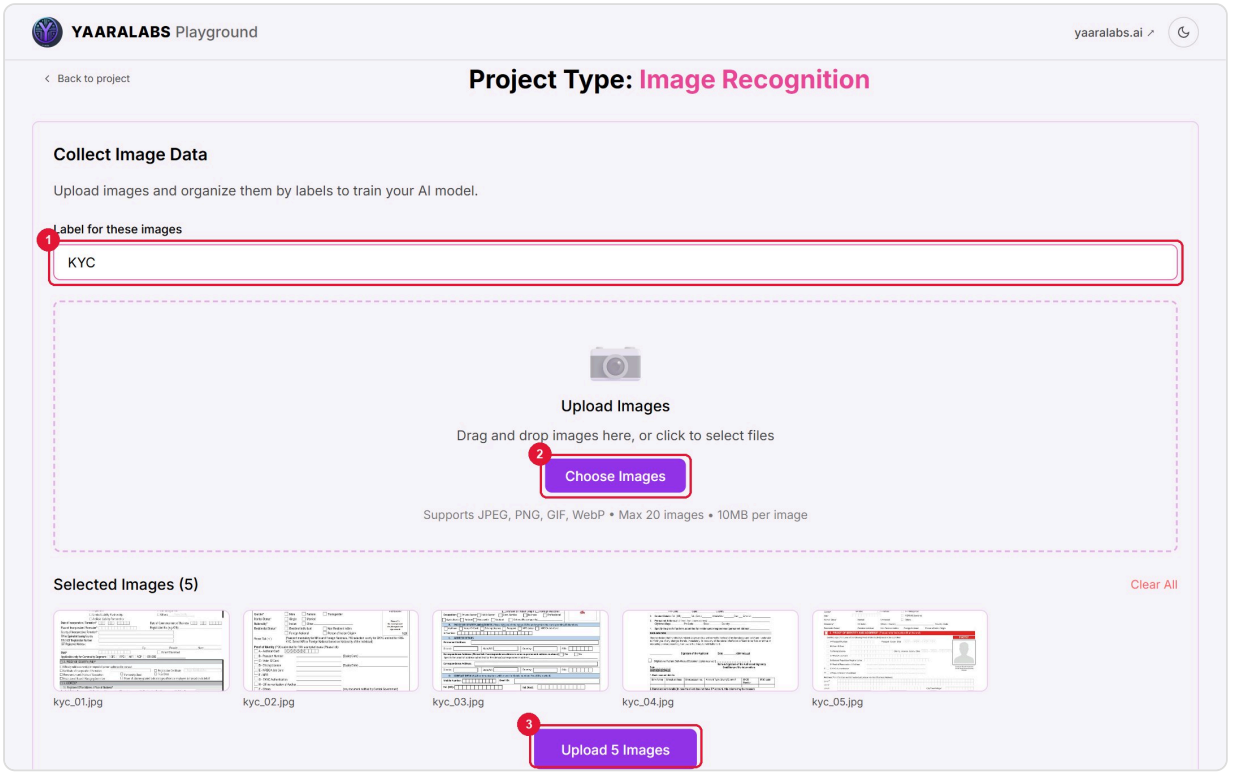
1 Create the project

Type a project name, choose “Image Recognition”, then click Create Project.



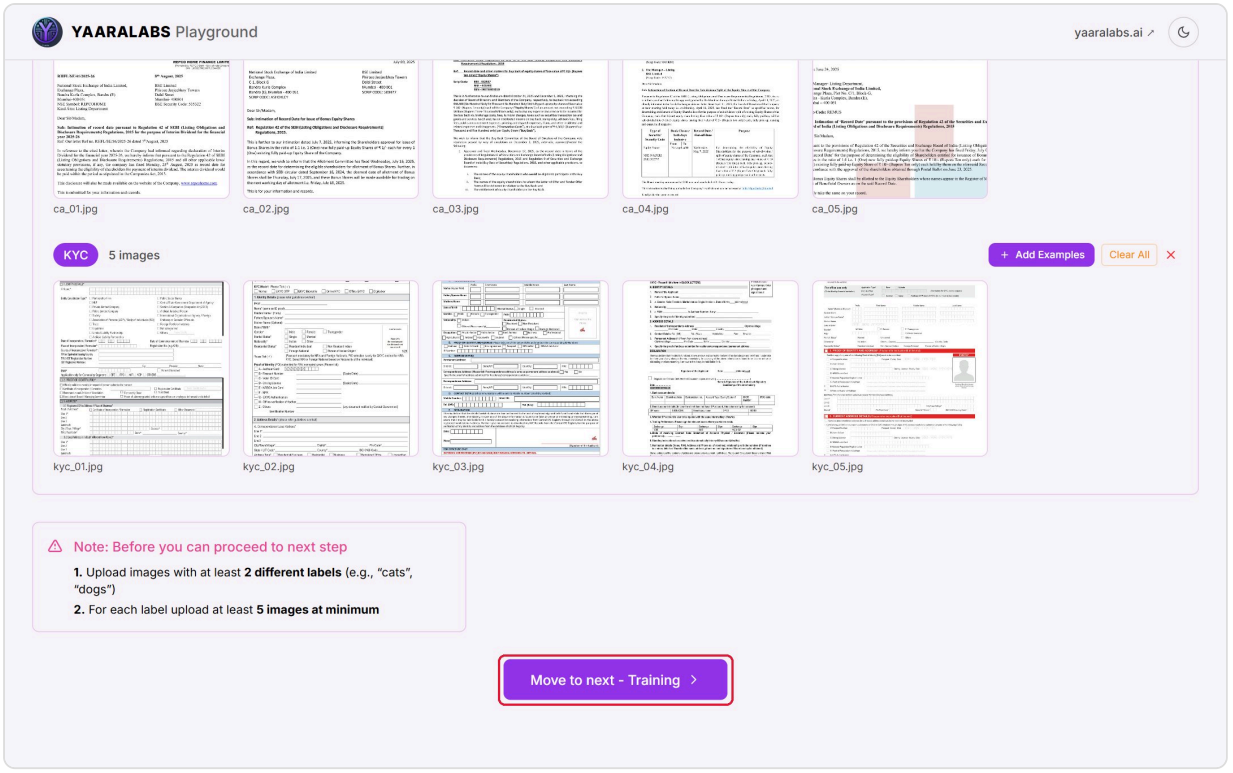
2 Add the first label and its images

Type the label (KYC), click Choose Images and pick the 5 KYC form images, then click Upload.



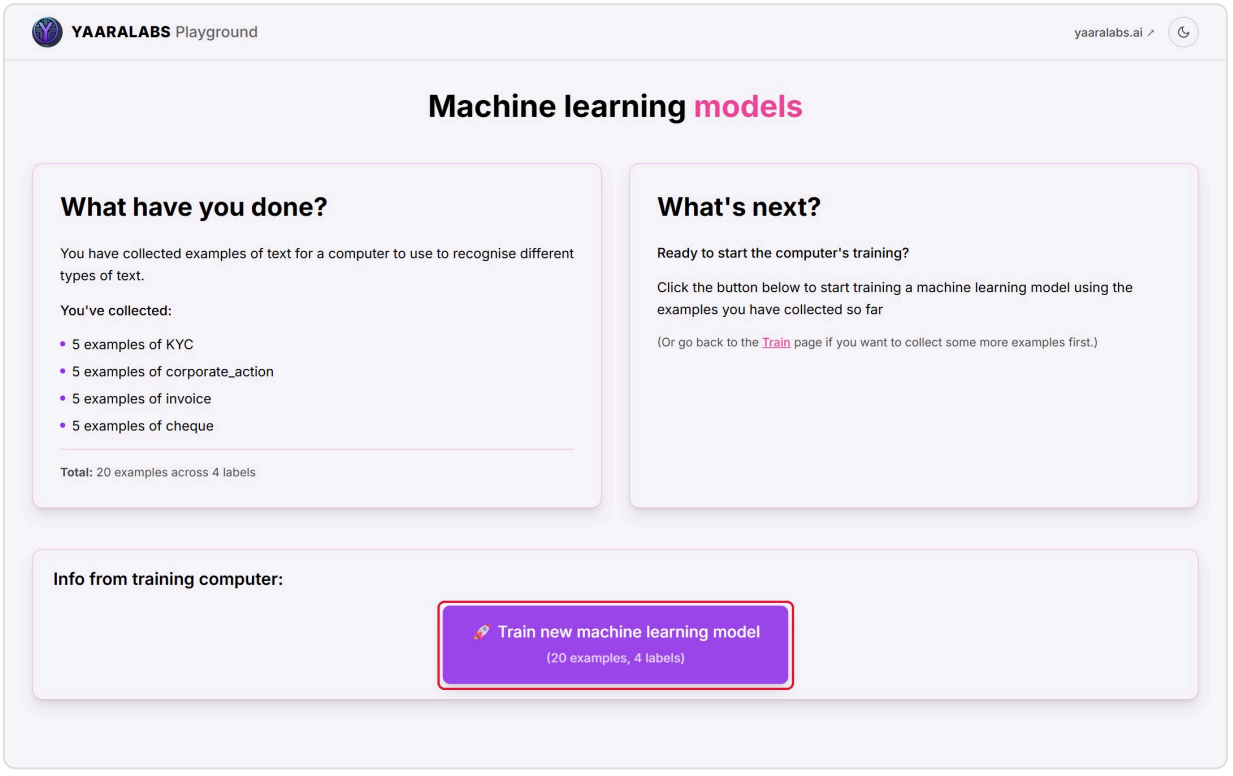
3 All four document types uploaded

Repeat for corporate_action, invoice and cheque: 4 labels × 5 images. Then Move to next – Training.



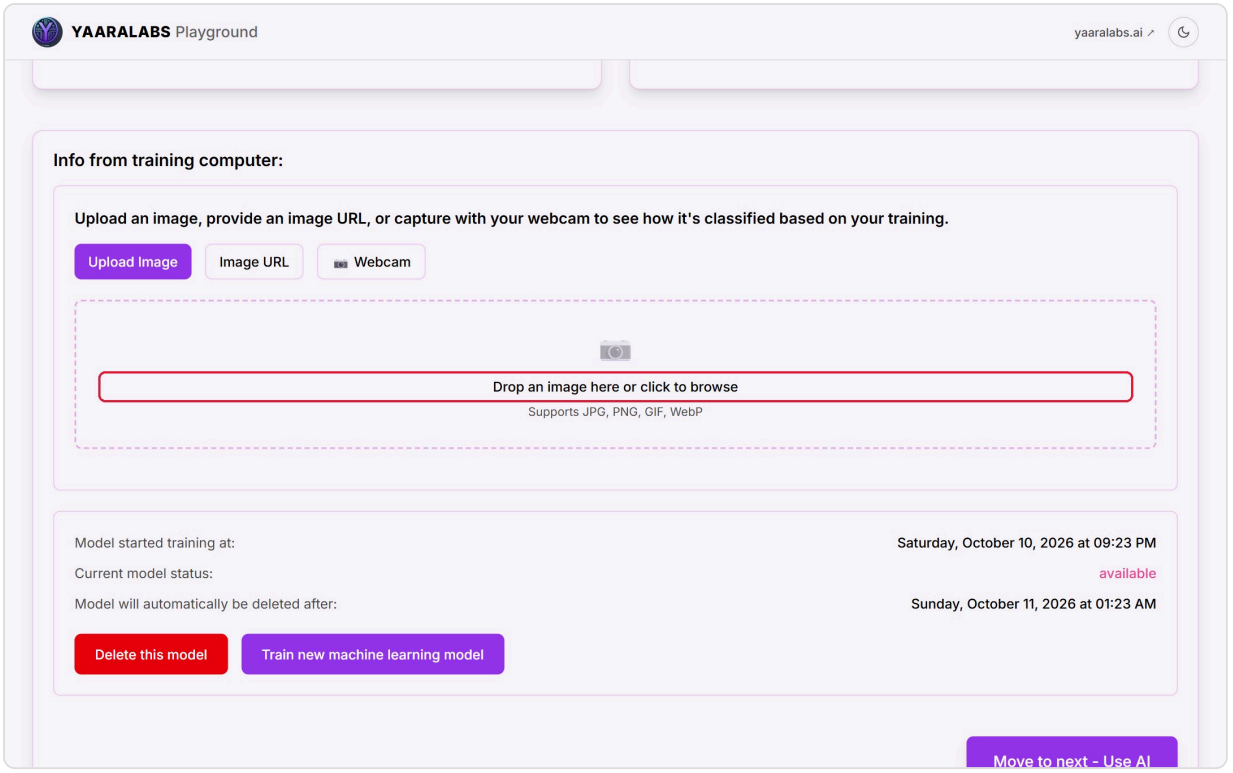
4 Train the image model

The page confirms 20 images across 4 labels. Click Train: MobileNetV2 transfer learning trains on the server.



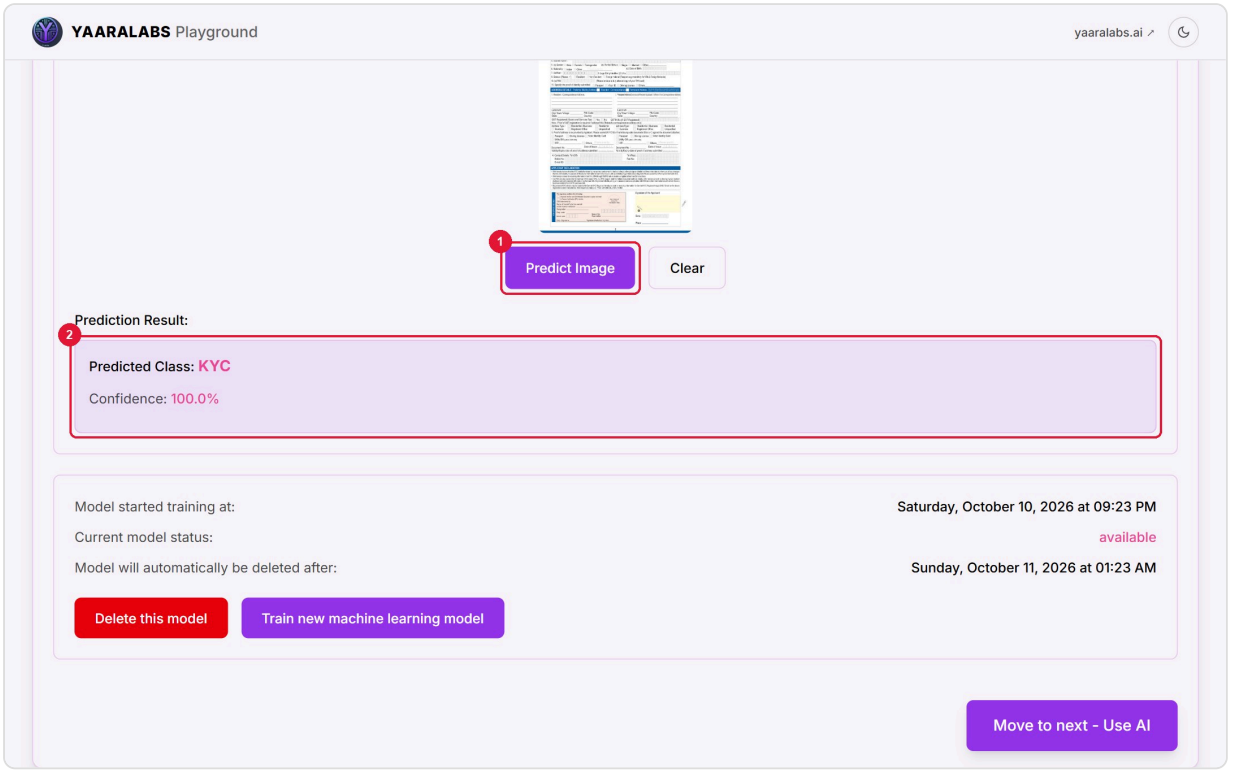
5 The model is ready

Training is done. Upload any document image here to see what the model thinks it is.



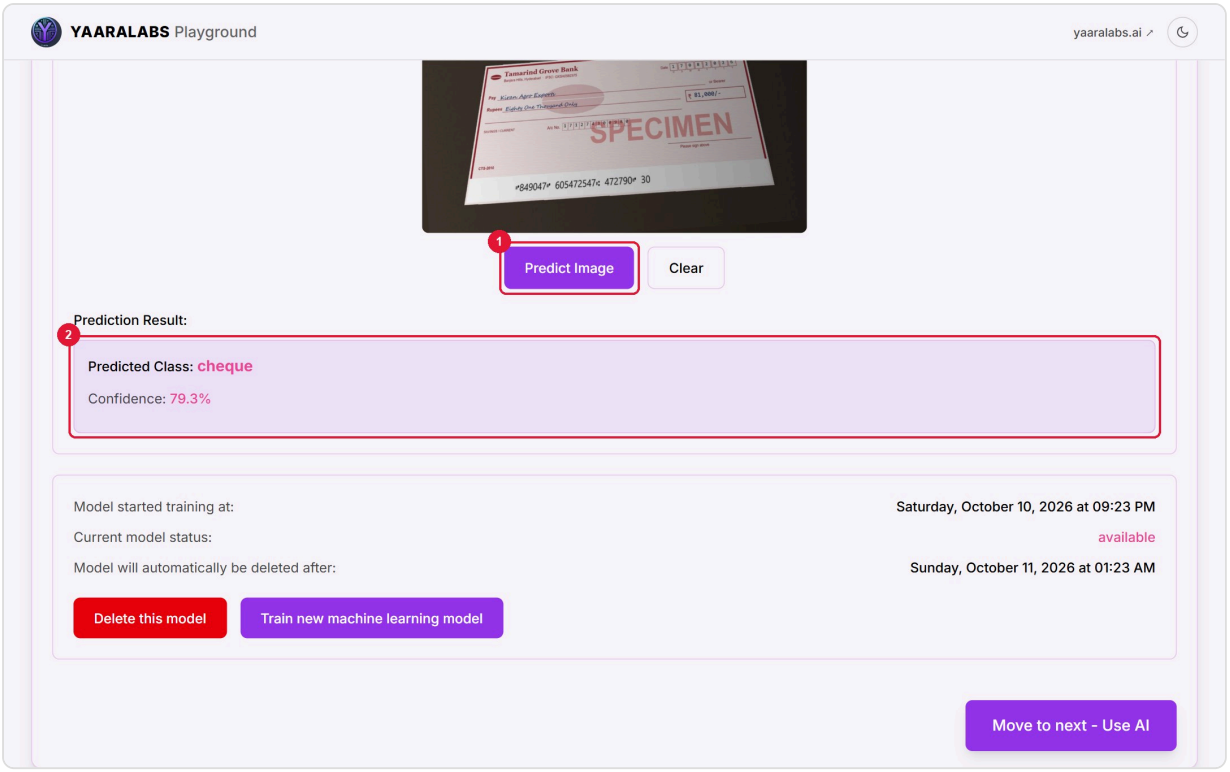
6 Test: KYC (clean)

test_01.jpg: a clean KYC image. Click Predict Image; the result shows the predicted class and confidence.



7 Test: cheque (phone-photo look)

test_05.jpg: a phone-photo look cheque image. Click Predict Image; the result shows the predicted class and confidence.



Results on the 12 test images

Overall	Clean images	Degraded images
10 of 12 correct	**4 of 4** correct	**6 of 8** correct

Test image	File and condition	Actually	Model said	Confidence	Result
	**test_01.jpg** clean: new document, upright	KYC	KYC	100.0%	**Correct**
	**test_02.jpg** clean: new document, upright	Invoice	Invoice	66.1%	**Correct**
	**test_03.jpg** partial crop: bottom 25% cut off	KYC	KYC	99.8%	**Correct**

	**test_04.jpg** clean: new document, upright	Corporate action	Corporate action	99.6%	**Correct**
	**test_05.jpg** phone-photo look: perspective skew 9 deg, rotation 4 deg, shadow gradient, on a dark desk	Cheque	Cheque	79.3%	**Correct**
	**test_06.jpg** Gaussian blur, radius 3	Corporate action	Corporate action	96.0%	**Correct**
	**test_07.jpg** heavy JPEG compression: quality 6	Invoice	Cheque	50.7%	**Wrong**
	**test_08.jpg** partial crop: right 22% cut off	Cheque	Cheque	100.0%	**Correct**
	**test_09.jpg** clean: new document, upright	Cheque	Cheque	100.0%	**Correct**
	**test_10.jpg** motion blur: 15 px horizontal	Invoice	Invoice	98.8%	**Correct**
	**test_11.jpg** low light / low contrast: brightness x0.45, contrast x0.5	Corporate action	Corporate action	71.3%	**Correct**
	**test_12.jpg** phone-photo look: perspective skew 12 deg, rotation -3 deg, shadow gradient, on a dark desk	KYC	Corporate action	69.5%	**Wrong**

Conclusion

WHAT THE EXPERIMENT SHOWS

- With only 20 training images, the model recognises document types it has never seen.
- Per type: KYC 2/3, Corporate action 3/3, Invoice 2/3, Cheque 3/3.

WHAT TO WATCH

- Mistakes: Invoice (heavy JPEG compression) read as Cheque; KYC (phone-photo look) read as Corporate action.

- Clean scans are the easy case; blur, low light, angled phone photos and crops are where it is tested.

- More and more varied training images (photos, scans, different issuers) make it sturdier.
- A low confidence score is the signal to send the document to a person.

Takeaway: transfer learning lets Nuvama build a useful document sorter from a handful of examples: 10 of 12 unseen test documents were recognised, including 6 of 8 deliberately degraded ones.